



NCIE-003

Data Model & Data Dictionary

In development — for human review — Version 0.2

Defines how the information required by NCIE-002's architecture is represented, related, governed and persisted

Authoritative canonical data reference for NCIE-004 through NCIE-018 and Codex implementation

Traces to NCIE-001 Master Product Requirements Document v1.2 (requirements authority) and NCIE-002 System Architecture & Technical Design v0.4 (architecture authority)

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Document Governance

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Owner	National Communications Intelligence Ecosystem (NCIE) Architecture Function
Reviewer	David King Boison (PhD) — Academic Intelligence Center
Status	IN DEVELOPMENT — FOR HUMAN REVIEW until explicitly approved, chapter by chapter

0.1 Authority Chain

NCIE-001 establishes what NCIE requires. NCIE-002 establishes how the system is architecturally structured. NCIE-003 establishes how the information required by that architecture is represented, related, governed and persisted. NCIE-003 implements NCIE-002 semantically; it does not reinterpret it. Where this document appears to require an architectural change to NCIE-002, the affected chapter flags it as **UPSTREAM ARCHITECTURE IMPACT — REVIEW REQUIRED** rather than silently modifying the architecture through the data model.

0.2 Baseline and Numbering Status

The NCIE-003 Skeletal v0.2 43-chapter structure is the approved structural baseline for this expansion; its original “FOR REVIEW — NOT YET APPROVED FOR FULL EXPANSION” stamp is superseded by that approval. The structure is locked unless a genuine architectural contradiction or material omission is discovered during expansion, in which case it is flagged rather than silently corrected. Separately, the NCIE Production Documentation Suite register identifies NCIE-003 as “Technology Stack & Engineering Blueprint”; that entry is stale relative to this approved artifact and is not overridden or renumbered by this document — reconciling the register, and deciding what becomes of the displaced document, is a documentation-governance decision outside this expansion's scope.

0.3 Chapter Status Convention

Every chapter in this edition is stamped **IN DEVELOPMENT — FOR HUMAN REVIEW**. Approval of the skeletal baseline authorizes expansion; it does not automatically approve every expanded interpretation. Chapters move to **Approved** only on explicit confirmation.

0.4 Human Review Focus Classification

Every chapter closes with a classified Human Review Focus register. Each item carries one of five labels:

- RESOLVED BY APPROVED NCIE BASELINE — already established by NCIE-001/002 or an approved NCIE-003 decision.
- PROPOSED DESIGN DEFAULT — technically recommended for the implementation baseline but subject to NCA confirmation.
- INSTITUTIONAL CONFIRMATION REQUIRED — cannot legitimately be settled from architecture alone.
- SOURCE/DATA DISCOVERY REQUIRED — depends on actual NCA/operator systems, schemas, fields, workflows or data characteristics not yet inspected.
- LEGAL/POLICY CONFIRMATION REQUIRED — depends on applicable NCA policy, records requirements, privacy, legal authority, sovereignty or regulatory interpretation.

The last three categories are never converted into invented facts to make the document appear complete. Chapter 42 consolidates every still-open item across all chapters into the authoritative **Open Institutional Decisions & Human Review Register**.

0.5 Depth Standard

Architecture already established in NCIE-002 is referenced, not rewritten. Each substantive data-model capability addresses, where relevant: purpose, semantic definition, principal entities, attributes/attribute classes, relationships and cardinality, identifiers, lifecycle/states, temporal semantics, provenance, validation/constraints, classification/security, ownership, failure/exception semantics, ARGUS implications where relevant, NCIE-002 traceability, human review requirements and acceptance criteria. Atomic definitions or invariants remain short where additional prose would add no technical value.

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Document Governance, Purpose & Data-Modelling Mandate

Defines why NCIE-003 exists, what it governs, and how its data definitions become authoritative across implementation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

1.1 Purpose

NCIE-003 is the semantic authority for NCIE's data: it defines what every canonical object means, how it is identified, how it relates to other objects, how it changes over time, where it came from, who owns it, how it is protected, and how downstream systems may consume it without changing its meaning. It governs after NCIE-001 (requirements) and NCIE-002 (architecture), and before NCIE-004 (implementation-level data engineering) and any physical database design.

1.2 Semantic Definition: Canonical, Authoritative, Source of Truth

Three terms recur throughout this document and are fixed here so later chapters do not redefine them locally:

- Canonical — the one NCIE-native representation of an entity or fact, independent of any single source system's format.
- Authoritative — the specific service or store that NCIE treats as correct for a given canonical object, per NCIE-002's domain-service ownership (e.g. the Rule Service is authoritative for rule values, per NCIE-002 Ch.22).
- Source of truth — the originating system or record from which a canonical fact was derived; distinct from authoritative, since NCIE can be authoritative for the canonical representation while the source system remains the source of truth for the original observation.

1.3 Principal Entities

Table 1. Chapter 1 governance entities.

Entity	Description
Document Metadata	Version, edition, status, prepared/approved dates, custodian.
Scope Statement	What NCIE-003 governs and explicitly excludes.
Normative Term	A defined term whose meaning is binding across all chapters (e.g. canonical, authoritative).
Change Record	A controlled, versioned change to a canonical definition.
Baseline Status	Per-chapter approval state (IN DEVELOPMENT — FOR HUMAN REVIEW / APPROVED).
Cross-Document Reference	A pointer to the specific NCIE-001/002 requirement or architecture element a definition traces to.

1.4 Change Control

A canonical definition change follows: proposal → impact assessment against dependent chapters and NCIE-002 traceability → approval by the confirmed authority (§1's Human Review Focus) → version increment → change record retained permanently. No canonical definition is edited in place without a retained record of its prior form, consistent with NCIE-002's evidence/provenance-first principle (NCIE-002 Ch.1 §1.3.3).

1.5 Ownership

NCIE-003 as a whole is owned by the NCIE Architecture Function. Individual chapters may name a domain steward (e.g. the Revenue Intelligence data model is stewarded jointly with Revenue Assurance) without transferring change-control authority away from the Architecture Function.

NCIE-002 Architecture Dependency & Traceability

NCIE-003 exists because NCIE-002 Ch.1 §1.3.3 (evidence/provenance-first) and Ch.5 (Canonical Data Platform) require canonical data whose meaning is fixed independently of any one domain service's implementation. This chapter is the governance root for that requirement.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the approval authority for NCIE-003 as a whole and for individual chapter sign-off.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm whether any existing NCA enterprise data-governance standard already exists and must be adopted rather than superseded by this chapter's change-control model.
- **[RESOLVED BY APPROVED NCIE BASELINE]** When a skeletal definition becomes an authoritative baseline: resolved by §0.2/§0.3 of this edition's front matter — skeletal-structure approval authorizes expansion; each chapter separately requires explicit approval before its content is binding.

1.6 Acceptance Criteria

- Every subsequent NCIE-003 chapter states its approval status using the convention fixed in §0.3.
- Every canonical definition change is demonstrably traceable to a Change Record with prior version retained.

Data Architecture Principles & Modelling Standards

Sets the modelling rules every NCIE domain must follow so that data remains consistent, temporal, traceable and secure.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

2.1 Purpose

Fixes the modelling rules every domain chapter in this document must follow, so canonical data remains consistent, temporal, traceable and secure across 43 chapters authored at different times by different contributors.

2.2 Semantic Definition

A modelling standard in this chapter is binding: a domain chapter that deviates from it must say so explicitly and justify the deviation, rather than silently using a different convention.

2.3 Canonical Modelling Rules

- Unknown, Partial, Stale and Unavailable are modelled as explicit, distinguishable states on every canonical fact — never collapsed into a null that could also mean zero or false (directly implements NCIE-002 Ch.5 §5.3's missing-is-not-zero principle and Ch.18 §18.5's National Operating Picture state vocabulary).
- Canonical identity never depends on a display name, human-friendly label, or user-facing product name (this is the data-model counterpart of NCIE-002 Ch.19 §19.11's ARGUS naming boundary, generalised to every entity, not only the Assistant).
- Provenance, time and classification are first-class properties on every canonical entity, not optional metadata bolted on later.

2.4 Naming Conventions

- Canonical entity names are singular, domain-neutral nouns (e.g. Incident, not Incidents or NetworkIncident).
- A field name never encodes a specific source system (e.g. operator_export_col_14 is prohibited as a canonical field name; it belongs only in a Source-System Field Mapping, §30, §39).
- Abbreviations are only used where already fixed by NCIE-001/002 (e.g. PM, KPI, SIM, MoMo, REWS, NOP, ARGUS).

2.5 Null / Unknown Semantics

Table 2. Explicit data-quality state vocabulary, binding across all domain chapters.

State	Meaning	Prohibited Substitute
Unknown	No observation exists yet for this fact.	Never represented as 0, empty string, or false.
Partial	An observation exists but does not cover the full expected scope (e.g. some cells missing from a district PM file).	Never silently treated as complete.
Stale	An observation exists but has exceeded its freshness window (NCIE-002 Ch.18 §18.5).	Never silently treated as current.
Unavailable	The source or pipeline that would produce this fact is currently degraded (NCIE-002 Ch.6, Ch.27).	Never conflated with Unknown, which is about the fact, not the pipeline.

2.6 Temporal Modelling Rules

Every canonical fact capable of changing over time carries, at minimum, occurrence time and knowledge time as distinct properties (full bitemporal model in Chapter 5). Rules and relationships that can change under governance carry effective-from/effective-to dates (Chapter 24).

2.7 Provenance Requirements

Every canonical fact references: the Source that supplied it (Chapter 6), the Ingestion Run that acquired it, and — where derived — the transformation chain that produced it (Chapter 25). No canonical fact may exist without at least one of these unless it is itself a foundational reference-data seed explicitly marked as such.

2.8 Extensibility and Versioning

- New attributes may be added to a canonical entity without a breaking version change, provided existing consumers are unaffected (additive-first, consistent with NCIE-002 Ch.25 §25.4).
- Removing or redefining the meaning of an existing attribute requires a new major schema version and a migration/deprecation plan (Chapter 33, Chapter 34).

2.9 Classification and Security

Every canonical entity declares a default classification tier (public/internal/protected/sensitive) at the entity level, which individual attributes may raise but never lower, consistent with NCIE-002 Ch.26 §26.3.

2.10 Failure / Exception Semantics

A domain chapter that cannot populate a required modelling property (e.g. no clear knowledge time available from a source) must model that gap explicitly as Unknown/Unavailable rather than omit the property, so downstream consumers can detect the gap programmatically.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.1 §1.3 (design principles), Ch.5 (Canonical Data Platform, layered architecture and temporal model), Ch.18 §18.5 (National Operating Picture state vocabulary) and Ch.26 (classification tiers).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Approve the enterprise naming conventions in §2.4 as binding across all NCIE domains, including any existing NCA-wide systems this may eventually touch.
- **[PROPOSED DESIGN DEFAULT]** The null/unknown state vocabulary in §2.5 is proposed as the binding standard for all 43 chapters; it is technically defensible from NCIE-002 alone and does not depend on undiscovered NCA data.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm whether NCA already operates an enterprise data-modelling standard (naming, null-handling or otherwise) that NCIE-003 must adopt rather than establish independently.

2.11 Acceptance Criteria

- Every domain chapter (Ch.7 onward) is checked against §2.3-§2.9 before being marked ready for review.
- A sampled canonical entity from at least three different domains demonstrably carries provenance, temporal and classification properties per this chapter's rules.

Enterprise Canonical Data Model

Provides the top-level view of the major objects and relationships across the whole NCIE ecosystem.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

3.1 Purpose

Provides the top-level, whole-of-NCIE view of major object families and how they relate, before any single domain chapter goes deep. This is the map that Chapters 7 onward each occupy one region of.

3.2 Semantic Definition

An enterprise entity is shared or referenced across more than one domain (e.g. Operator, Evidence); a domain entity belongs to exactly one domain chapter's ownership (e.g. PM Observation belongs to Chapter 7).

3.3 Principal Entity Families

Table 3. Enterprise entity families and their owning domain chapters.

Family	Representative Entities	Owning Chapter(s)
Shared / Core	Operator, Geography, Cell, Identity Reference, Source, Evidence	Ch.3, Ch.4, Ch.6, Ch.8, Ch.17
Domain — Network & Field	PM Observation, Topology Relation, Campaign, Measurement	Ch.7, Ch.8, Ch.9
Domain — Commercial	Traffic Observation, Revenue Calculation, Mobile Money Aggregate	Ch.11, Ch.12, Ch.13
Domain — Identity & Fraud	Registration, SIMBOX Detection, Blocking Action	Ch.14, Ch.15
Regulatory	Case, Finding, Decision	Ch.16
Analytical	Condition, Alert, Situation, Graph Edge	Ch.18, Ch.19
Collaborative	Room, Hypothesis, ARGUS Interaction, Memory Object	Ch.20, Ch.21, Ch.22
Platform	Workflow Instance, Rule, Audit Event, Event Envelope	Ch.23, Ch.24, Ch.27, Ch.28

3.4 Cross-Domain Relationship Principle

A domain entity references a shared entity by canonical reference (Chapter 4), never by copying the shared entity's attributes inline. A Traffic Observation references Operator by ID; it does not carry its own duplicate operator-name field that could drift out of sync.

3.5 Enterprise Relationship Diagram

Figure 1. Enterprise-level relationships between entity families.

SOURCE	RELATIONSHIP	TARGET
Shared/Core entities	are referenced by	Domain entities (Network, Commercial, Identity & Fraud)
Domain entities	are correlated by	Cross-Domain Fusion & Intelligence Graph (Ch.19)
Domain entities + Fusion	become candidates for	Regulatory Case, Finding & Decision (Ch.16)
All entities	carry references into	Evidence & Provenance (Ch.17, Ch.25)
Humans + ARGUS	reason over entities within	Collaborative Intelligence (Ch.20, Ch.21)

3.6 Ownership Boundaries

Each entity has exactly one owning chapter for its canonical definition. Other chapters may extend it with domain-specific attributes only through an explicit extension relationship, never by redefining the shared entity's core meaning.

3.7 Semantic Separations (Enterprise Level)

This chapter is where the enterprise-wide semantic separations mandated for NCIE-003 are first declared as entity-family boundaries, not merely explanatory prose:

- Source Data ≠ Evidence (Ch.6 vs Ch.17).
- Observation ≠ Condition ≠ Alert ≠ Situation (Ch.7/Ch.11 vs Ch.18).
- Hypothesis ≠ Finding ≠ Decision (Ch.20/Ch.21 vs Ch.16).
- Detection ≠ Confirmed Fraud, and Registered Identity ≠ Fraud Actor (Ch.15).
- Requested ≠ Approved ≠ Executed ≠ Verified (Ch.23).
- Memory ≠ Current Truth, and Institutional Knowledge ≠ Rule (Ch.22 vs Ch.24).
- ARGUS Contribution ≠ Evidence, and Graph Relationship ≠ Causality (Ch.20, Ch.19).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.5 (Canonical Data Platform, canonical entity registry) and Ch.17 (Cross-Domain Fusion correlation-not-causation principle).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the principal enterprise entities in §3.3 against NCA's actual organisational view of its own data assets.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Identify any NCA-wide master entity not yet represented (e.g. an existing enterprise operator master list) that should be adopted as the canonical source rather than re-created.

3.8 Acceptance Criteria

- Every entity introduced in Chapters 7-28 is traceable to exactly one family in §3.3 and one owning chapter.
- None of the semantic separations in §3.7 is violated by a foreign-key structure that would allow two distinct states to merge (verified per-domain in each owning chapter and consolidated in Ch.40).

Common Entity, Identifier & Reference Model

Defines how NCIE identifies the same real-world or system object consistently across domains and source systems.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

4.1 Purpose

Defines how NCIE identifies the same real-world or system object consistently across domains and source systems, so that a cell, operator or identity referenced in five different chapters is provably the same object each time.

4.2 Semantic Definition

- Canonical ID — the stable NCIE-native identifier for an entity, assigned once and never reused, surviving renames.
- External/Source ID — the identifier a specific source system uses for the same real-world object; retained for lineage, never used as the canonical key.
- Alias — a human-facing alternative label (including a configurable presentation name such as ARGUS for the Assistant service identity) that never substitutes for canonical identity.

4.3 Principal Entities

Table 4. Common Entity, Identifier & Reference Model principal entities.

Entity	Attributes / Attribute Classes
Canonical ID	entity type, id value, created date, status (active/merged/retired).
External/Source ID	canonical ID reference, source reference (Ch.6), source-native id value, first/last observed.
Alias	canonical ID reference, alias value, alias type (display name, legacy name, presentation identity), effective period.
Identity-Resolution State	candidate match set, confidence, resolution status (resolved/ambiguous/rejected).
Merge/Split Lineage	prior canonical ID(s), resulting canonical ID(s), reason, actor, timestamp.

4.4 Identity Resolution Rules

- An ambiguous match (more than one plausible canonical entity) is stored as an Identity-Resolution State with status ambiguous and is never force-resolved automatically to the highest-confidence candidate without a rule or human decision permitting it (directly implements NCIE-002 Ch.14 §14.6, Ch.34 §34.1's ambiguous-match handling generalised platform-wide).
- A canonical ID is never reused after retirement, even if the original entity is later found to be identical to a new one — reconciliation produces a Merge/Split Lineage record instead.

4.5 Lifecycle / States

Figure 2. Identity resolution lifecycle from first observation to canonical assignment.

01	New object observed with no confident canonical match
02	Identity-Resolution State created (candidate set, confidence)
03	Resolved automatically (high confidence per governed threshold) or by human decision
04	Canonical ID assigned or merged into existing canonical ID

4.6 Relationships and Cardinality

- One Canonical ID has zero-to-many External/Source IDs (one per source system that has independently referenced it).
- One Canonical ID has zero-to-many Aliases, at most one active per alias type at a time (effective-dated, Chapter 24 pattern).
- A Merge/Split Lineage record has one-to-many prior canonical IDs and one-to-many resulting canonical IDs, supporting both merges and splits.

4.7 Provenance and Classification

Every External/Source ID retains which Source (Chapter 6) supplied it and when, for lineage. Identifier records themselves inherit the classification of the entity they identify — a protected-identity canonical ID (Chapter 14) is itself protected metadata, not merely the attributes attached to it.

4.8 Failure / Exception Semantics

A source supplying an identifier that cannot be resolved to any canonical entity, and does not meet the bar for creating a new one, is retained as an unresolved External/Source ID with a flagged exception state rather than silently discarded — this preserves the option to resolve it later without re-ingesting the original source data.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.5 §5.2 (canonical entity registry), Ch.17 (entity resolution for fusion) and Ch.19 §19.11 (ARGUS as a presentation alias over a stable internal identity).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm identifier formats where institutionally constrained (e.g. any existing NCA-mandated operator or facility identifier scheme).
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm authoritative master-data sources for core entities (Operator, Geography) — is there an existing NCA master list, or does NCIE-003 establish the first canonical one?
- **[PROPOSED DESIGN DEFAULT]** The ambiguous-match handling in §4.4 (never force-resolve automatically) is proposed as the binding default for all identity resolution across NCIE-003, extending the pattern already fixed for SIM/identity matching in NCIE-002 Ch.14.

4.9 Acceptance Criteria

- A simulated rename of a real-world entity (e.g. a cell relocated and renamed) is demonstrably resolved without losing its Canonical ID or historical External/Source ID linkage.
- An ambiguous identity match is demonstrably retained as ambiguous rather than silently merged.

Temporal & Bitemporal Data Model

Defines all time semantics needed to reconstruct what happened, when NCIE learned it, and which rules applied.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

5.1 Purpose

Defines every time semantic needed to reconstruct what happened, when NCIE learned it, and which rules applied — the data-model foundation for NCIE-002's decision-time reconstruction requirement (NCIE-002 Ch.23 §23.5).

5.2 Semantic Definition

Table 5. NCIE-003 time-type vocabulary, binding across all chapters.

Time Type	Definition
Occurrence/Observation Time	When the real-world fact actually happened or was measured.
Acquisition Time	When NCIE's ingestion pipeline received the artifact containing the fact (Ch.6).
Knowledge Time	When NCIE considered the fact known/validated — may lag acquisition time and must never be back-dated to occurrence time for late-arriving data.
Effective-From / Effective-To	The period during which a rule, relationship or configuration value is in force (Ch.24).
Decision Time	The point in time as of which a Decision (Ch.16) is evaluated, excluding any evidence that arrived later (NCIE-002 Ch.23 §23.5).
Revision Time	When a correction to a previously recorded fact was made (distinct from the fact's own occurrence time).
Scenario Time	A hypothetical reference time used by a what-if Scenario (Ch.24 §24 pattern); never written to production history.

5.3 Bitemporal Storage Principle

An entity requiring historical defensibility stores both occurrence/effective time and knowledge/revision time as independent axes: querying "as it was known on date X" and querying "as it occurred/applied on date Y" must be answerable independently and in combination, without one overwriting the other.

5.4 Principal Entities

- Temporal Fact Envelope — the reusable structure (occurrence time, acquisition time, knowledge time, revision time, superseded-by reference) that domain entities requiring bitemporal storage embed.
- Historical Snapshot — a reconstructed view of canonical state as of a specified knowledge time (implements NCIE-002 Ch.23's Historical Snapshot Service data requirements).
- Decision-Time Context — the bounded set of facts, rule versions and evidence available at a Decision's timestamp, explicitly excluding later arrivals.

5.5 Which Entities Require Bitemporal Storage

Full bitemporal storage (both axes independently queryable) is required for: canonical facts feeding Revenue calculation (Ch.12), Rules (Ch.24), Registration limits (Ch.14), Evidence (Ch.17), and Findings/Decisions (Ch.16). Simpler single-axis (knowledge-time-only) storage is acceptable for high-volume, low-materiality telemetry such as raw PM samples (Ch.7), provided the coarser Ingestion Run (Ch.6) retains acquisition time.

5.6 Correction Semantics

A correction to a previously recorded fact never overwrites the prior value in place. It creates a new version with its own revision time, linked to the prior version via a Correction relationship (Ch.25), so a query as of a knowledge time before the correction still returns the pre-correction value.

5.7 Failure / Exception Semantics

A source that cannot supply a clear occurrence time (only a delivery time) has that gap recorded explicitly — occurrence time is marked Unknown (Chapter 2's vocabulary) rather than defaulted to acquisition time, which would silently misrepresent when the fact actually happened.

5.8 Official Time Zone and Reference Clock

All canonical timestamps are stored in a single reference time zone (UTC recommended) with local Ghana time (GMT, no DST) derived for display, so cross-source comparison is never subject to ambiguous local-time arithmetic.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.5 §5.3 (temporal/metadata model), Ch.22 §22.3 (rule effective dating) and Ch.23 §23.5 (decision-time reconstruction).

Human Review Focus

- **[PROPOSED DESIGN DEFAULT]** UTC as the canonical storage time zone with Ghana local time derived for display is proposed as the binding default (§5.8); technically defensible and does not depend on undiscovered facts.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which entities beyond the §5.5 list require full bitemporal storage versus knowledge-time-only, weighing storage cost against legal/operational reconstruction needs.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm legal/operational expectations for how far back historical reconstruction must be defensible (informs retention in Ch.31 and archival tiering in Ch.35).

5.9 Acceptance Criteria

- A correction to a Revenue-relevant Traffic fact is demonstrably retrievable both in its pre- and post-correction form, keyed by knowledge time.
- A Decision-Time Context reconstruction for a sampled Case demonstrably excludes evidence with a knowledge time after the Decision's timestamp.

Source, Provider, Acquisition & Ingestion Data Model

Models where data comes from, how it enters NCIE, and how source lineage is retained.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

6.1 Purpose

Models where data comes from, how it enters NCIE, and how source lineage is retained — the data-model realisation of NCIE-002's Source Registry and Ingestion Gateway (NCIE-002 Ch.6).

6.2 Semantic Definition

- Source — the institutional origin of data (e.g. "MTN monthly PM export"), distinct from the technical channel used to acquire it.
- Provider — the technical acquisition channel (API, file, email, screenshot, manual, RPA) currently used for a Source; a Source may change Provider without changing identity.
- Multiple technical copies of the same underlying observation (e.g. the same file received via two channels) do not create independent corroboration — they are deduplicated to one Source Object Reference before being treated as evidence of anything.

6.3 Principal Entities

Table 6. Source, Provider, Acquisition & Ingestion principal entities.

Entity	Attributes / Attribute Classes
Source	name, owner, domain, expected frequency, trust tier, status.
Provider	type (API/file/email/screenshot/manual/RPA), source reference, configuration reference (credentials by reference only, per NCIE-002 Ch.26).
Acquisition Channel	provider reference, endpoint/mailbox/portal reference, authentication reference.
Ingestion Run	source reference, provider reference, start/end time, status, artifact reference(s).
Source Object Reference	ingestion run reference, raw artifact pointer, hash/integrity metadata, deduplication key.
Validation Result	source object reference, checks performed, pass/fail, rejection reason.
Transformation Reference	source object reference, transformation applied, resulting canonical fact reference(s) (Ch.25).
Acquisition Status	source reference, current health (Green/Amber/Grey per NCIE-002 Ch.6 §6.9), last successful run.

6.4 Lifecycle / States

Figure 3. Source-to-canonical acquisition lifecycle.

01	Source registered with owner, frequency and provider
02	Acquisition Channel configured (credentials by reference only)
03	Ingestion Run attempted
04	Source Object Reference created from received artifact
05	Validation Result recorded (pass/fail/reason)

	▼
06	Transformation Reference links validated content to canonical facts
	▼
07	Acquisition Status updated (Green/Amber/Grey)

6.5 Relationships and Cardinality

- One Source has one current Provider but may retain historical Provider records (e.g. manual-to-RPA transition) without losing Source identity.
- One Ingestion Run produces zero-to-many Source Object References (e.g. a single file yielding multiple records).
- One Source Object Reference has zero-or-one Validation Result and, if passed, one-to-many Transformation References.

6.6 Deduplication Rule

Two Source Object References with the same deduplication key (content hash plus source plus occurrence-time bucket) are treated as one substantive observation for corroboration purposes, even if acquired through different Providers or Ingestion Runs — this directly implements the multiple-technical-copies-are-not-independent-corroboration principle.

6.7 Failure / Exception Semantics

- A failed Validation Result quarantines the Source Object Reference; it never silently becomes canonical, and it never silently disappears — it remains queryable as a rejected artifact with its reason (NCIE-002 Ch.6 §6.4).
- A Source with no successful Ingestion Run within its expected frequency window shows Acquisition Status Grey, not a false Green.

6.8 Classification and Ownership

Every Source has a named accountable owner before being enabled (NCIE-002 Ch.6 §6.6). Source Object References inherit the classification of the content they carry once known; until validated, raw content defaults to the most protective applicable classification tier.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.6 in full (Source Registry, Ingestion Gateway, Request/Run/Artifact/Validation lineage, health thresholds).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Inventory actual NCA/operator source systems, their current owners and provider channels — this chapter's entities are structurally ready but not yet populated with real sources.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm source ownership and trust hierarchy where more than one plausible owner exists (e.g. a source jointly relevant to Network and Regulatory domains).
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm which sources arrive via API, file, stream, RPA or manual upload today, to correctly populate initial Provider records.

6.9 Acceptance Criteria

- A deliberately duplicated source artifact (same content, two channels) is demonstrably deduplicated to one Source Object Reference before downstream use.
- A rejected Validation Result is demonstrably retrievable with its rejection reason, distinct from data that never arrived at all.

Network Intelligence Data Model

Defines the canonical data objects for network performance monitoring and QoS intelligence.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

7.1 Purpose

Defines the canonical data objects for network performance monitoring (PM) and QoS intelligence, implementing NCIE-002 Ch.7's KPI Engine and Investigation Service.

7.2 Semantic Definition

- PM availability (whether expected files/records arrived, Green/Amber/Grey) is a distinct property from QoS interpretation (whether the KPI value itself is good or bad) — the two are never merged into one status.
- District-level degradation evidence never becomes cell-level evidence merely by geographic containment; a cell is only shown as degraded when cell-level PM data supports it (NCIE-002 Ch.7 §7.4).

7.3 Principal Entities

Table 7. Network Intelligence principal entities.

Entity	Attributes / Attribute Classes
Operator	canonical ID (Ch.4), name, licence reference, status.
Site	canonical ID, operator reference, location reference (Ch.8).
Cell	canonical ID, site reference, technology, topology version (Ch.8), status.
Technology	enumeration (2G/3G/4G/5G and successors), governed reference data.
KPI	name, formula reference (Ch.24), unit, effective definition version.
PM Observation	cell reference, KPI reference, date/period, value, source reference (Ch.6), coverage status.
Baseline	KPI reference, scope (cell/site/district), reference value, effective period.
Threshold	KPI reference, threshold value, severity, effective period (Ch.24).
Network Condition	scope, KPI(s) involved, detected state, evidence references — the Ch.18 Condition object as realised for the Network domain.

7.4 Lifecycle and Relationships

Figure 4. District-to-cell network investigation data flow.

01	PM Observation ingested per cell/KPI/period (Ch.6)
02	Coverage status computed (Green/Amber/Grey)
03	KPI value compared to Threshold/Baseline
04	District-level degradation detected
05	Topology resolution to constituent Cells (Ch.8)
06	Cell-level PM Observation confirms or refutes degradation
07	Network Condition raised only on cell-level evidence

7.5 Temporal Semantics

PM Observation carries occurrence period (the measured interval), acquisition time and knowledge time (Ch.5). Baseline and Threshold are effective-dated (Ch.24) so a KPI's definition of "good" can change over time without corrupting historical evaluations.

7.6 Validation and Classification

- A PM Observation failing source/date/scope validation (Ch.6) is quarantined, never silently included in KPI aggregation.
- Network Condition and PM Observation are internal/operational classification by default; no protected-identity data is modelled in this chapter.

7.7 Failure / Exception Semantics

A cell with no PM Observation for the period is modelled as coverage status Grey with KPI value Unknown (Ch.2 vocabulary) — never defaulted to a passing or failing KPI value.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.7 in full (NMS Adapters, PM Service, KPI Engine, Investigation Service, coverage-status model).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm available network source fields per operator NMS/PM export format.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the authoritative KPI set, formulas and units (Ch.24 Rule Service will hold the governed values once confirmed).
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm cell/site master-data authority and update process — is there an existing NCA or operator-supplied cell master list?

7.8 Acceptance Criteria

- A synthetic district degradation with no cell-level PM data does not produce a Network Condition; the same scenario with confirming cell-level PM Observations does.
- A cell missing PM data for a period shows Grey/Unknown, never a false passing KPI.

Topology, GIS & Spatial Data Model

Defines the geographic and network-topology objects used for mapping and spatial analysis.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

8.1 Purpose

Defines the geographic and network-topology objects used for mapping, spatial analysis and the Madina-style place-to-cell resolution flow (NCIE-002 Ch.8 §8.3).

8.2 Semantic Definition

- Spatial precision is an explicit attribute of every location fact, not an assumed property of the coordinate value's apparent decimal length.
- National/operator-only aggregate data must never be placed on a map at a precision it does not actually possess.
- Anti-Fraud location evidence (Ch.15) is a distinct, more strongly controlled data class than ordinary network/topology location.

8.3 Principal Entities

Table 8. Topology, GIS & Spatial principal entities.

Entity	Attributes / Attribute Classes
Region / District	canonical ID (Ch.4), name, parent geography reference, boundary version.
Site Geometry	site reference (Ch.7), coordinates, coordinate reference system, precision.
Cell Location	cell reference (Ch.7), geometry, sector/azimuth where applicable, precision.
Coverage Area	cell/site reference, derived polygon, method, effective period.
Coordinate Reference System	governed reference data; the single system canonical geometry is stored in.
Spatial Precision	enumeration (exact/approximate/aggregate-only), applies to any location fact.
Topology Relation	cell-to-geography and cell-to-site relationships, versioned (Ch.5).

8.4 Temporal Semantics

Topology Relation and Coverage Area are versioned by effective period so historical topology ("what cells covered this district as of date X") is reconstructable, per NCIE-002 Ch.8's historical-topology requirement.

8.5 Place-to-Cell Resolution Data Flow

Figure 5. Place-to-cell resolution supported by the topology data model.

SOURCE	RELATIONSHIP	TARGET
Natural-language place name	resolves to	Region/District (canonical geography)
Region/District	resolves via Topology Relation to	constituent Cells
Cells	supply	PM Observation (Ch.7)

8.6 Classification and Security

Cell Location and Coverage Area default to internal classification. Any location fact associated with Anti-Fraud investigation (Ch.15) is reclassified sensitive and is never joined into a general-purpose map layer query without passing through Ch.15's sensitive-join controls.

8.7 Failure / Exception Semantics

An ambiguous place-name resolution (matching more than one Region/District candidate) is returned as a candidate set for human disambiguation, never silently resolved to the first match (NCIE-002 Ch.8 §8.6).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.8 in full (Topology Service, Spatial Database, Geocoder, Map API, historical topology, sensitive-layer permission filtering).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm authoritative GIS datasets (region/district boundaries, site coordinates) currently available at NCA/operators.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm coordinate system and precision standards to be used as the canonical storage CRS.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which users/roles may see sensitive location layers (cross-references NCIE-002 Ch.4 role model).

8.8 Acceptance Criteria

- A historical Topology Relation query returns the topology as it existed at a specified prior date.
- A location fact lacking real precision is never displayed at a finer precision than it actually has.

QoS Campaign & Drive-Test Data Model

Defines field-testing campaigns, routes, devices, test points and measurements.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

9.1 Purpose

Defines field-testing campaigns, routes, devices, test points and measurements, implementing NCIE-002 Ch.9's Campaign Repository and Matcher.

9.2 Semantic Definition

- Campaign metadata (operator, date, route, technology, method) is modelled separately from its Measurements, since metadata governs how measurements should be interpreted.
- Device, method and software/firmware version are preserved per campaign so measurement comparability can be judged rather than assumed.

9.3 Principal Entities

Table 9. QoS Campaign & Drive-Test principal entities.

Entity	Attributes / Attribute Classes
Campaign	canonical ID, operator reference, date range, route reference, technology, methodology reference, status.
Route	geometry, associated Region/District references (Ch.8).
Test Device	make/model, firmware/software version.
Test Point	campaign reference, geolocation (Ch.8 spatial precision), timestamp.
Measurement	test point reference, metric reference, value, unit.
Metric	governed reference data (e.g. RSRP, throughput, call-setup success).
Methodology	description, version, applicable metrics.
Campaign Status	coverage/completeness, evidential status.

9.4 Relationships and Geography Linkage

Measurements link to canonical Cell (Ch.7) and Region/District (Ch.8) references via the Matcher process (NCIE-002 Ch.9 §9.3), never via free-text location strings. Latest-applicable campaign resolution is geography-specific, per NCIE-002 Ch.9 §9.3 — a newer campaign in one district does not make an older, unrelated district's campaign stale by association.

9.5 Temporal and Provenance Semantics

Every Measurement retains campaign date, device and methodology, so age and comparability are always derivable, not assumed. Campaigns are never edited in place after ingestion; a correction is a new Campaign version with lineage to the original (Ch.25).

9.6 Classification

Campaign and Measurement data default to internal classification unless a specific campaign is flagged sensitive by its owning source agreement.

9.7 Failure / Exception Semantics

Route geometry that fails to match any known geography is retained as unmatched rather than discarded, pending manual topology reconciliation (mirrors NCIE-002 Ch.9 §9.6).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.9 in full (Campaign Repository, Geometry Store, Matcher, Analytics Service).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm existing drive-test systems, vendors and export formats.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm mandatory campaign metadata for a campaign to be treated as evidential.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm retention and evidential status of drive-test outputs for regulatory purposes.

9.8 Acceptance Criteria

- For a geography sampled by multiple campaigns over time, the latest-applicable campaign is correctly identified while all historical campaigns remain retrievable.
- Unmatched route geometry is demonstrably retained and flagged rather than dropped.

Incident Intelligence Data Model

Defines reported and observed incidents, affected scope, restoration and incident timelines.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

10.1 Purpose

Defines reported and observed incidents, affected scope, restoration and incident timelines, implementing NCIE-002 Ch.10's Mailbox Connector, Extractor, Threading Service and Incident Workflow.

10.2 Semantic Definition

- Operator-reported incident state and independently observed (NMS-corroborated) state are distinct properties on the same Incident, never merged into one status.
- Occurrence time (when the outage actually began) and reporting time (when the operator's email arrived) are both preserved.
- Restoration Reported (operator claims restored) is distinct from Restoration Verified (NMS evidence corroborates it); an Incident does not close on the former alone.

10.3 Principal Entities

Table 10. Incident Intelligence principal entities.

Entity	Attributes / Attribute Classes
Incident	canonical ID, category, occurrence time, reporting time, status, reported vs verified restoration.
Incident Report	incident reference, source email reference (Ch.6, Ch.17 evidence), extracted fields, thread reference.
Affected Operator	incident reference, operator reference (Ch.7).
Affected Network Object	incident reference, cell/site reference (Ch.7).
Affected Geography	incident reference, region/district reference (Ch.8).
Restoration Event	incident reference, reported time, verification method, verified time (nullable/Unknown).
Incident Update	incident reference, source report reference, update content, timestamp.
Incident Status	governed enumeration (open/monitoring/restoration-claimed/restoration-verified/closed).

10.4 Lifecycle

Figure 6. Incident lifecycle from report to verified closure.

01	Incident Report received (original email retained as Evidence, Ch.17)
02	Extractor produces structured fields
03	Threading Service links to existing Incident or opens new one
04	Restoration Reported recorded from operator update
05	Restoration Verified checked against Network Condition/PM Observation (Ch.7)
06	Incident closes only once verified, or remains open pending verification

10.5 Provenance

Every Incident Update retains its source Incident Report and, transitively, the original email evidence — the email itself is never modified, consistent with NCIE-002 Ch.10 §10.5.

10.6 Failure / Exception Semantics

A restoration claim unsupported by NMS evidence keeps the Incident in restoration-claimed status, not closed — closure requires either verification or an explicit, audited manual override with recorded justification.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.10 in full.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm incident reporting workflow and mailbox structure currently used by operators.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm operator incident email fields available for extraction.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm what constitutes validated restoration for closure purposes (i.e. which NMS evidence is sufficient).

10.7 Acceptance Criteria

- A multi-message incident thread is demonstrably consolidated into one Incident with full lineage to each source email.
- A restoration claim unsupported by NMS evidence is demonstrably kept open.

Traffic Intelligence Data Model

Defines international, off-net and on-net traffic observations and their reporting semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

11.1 Purpose

Defines International, Off-Net and On-Net traffic observations and their reporting semantics, feeding Revenue calculation (Ch.12) and cross-domain fusion (Ch.19).

11.2 Semantic Definition

Traffic type, direction, unit and reporting period are never implicit — every Traffic Observation states all four explicitly, since Revenue calculation correctness depends on them (NCIE-002 Ch.11 §11.2).

11.3 Principal Entities

Table 11. Traffic Intelligence principal entities.

Entity	Attributes / Attribute Classes
Traffic Observation	canonical ID, operator reference (Ch.7), traffic type, direction, period, volume, unit, source reference (Ch.6), coverage status.
Traffic Type	governed enumeration (International/Off-Net/On-Net).
Direction	governed enumeration (incoming/outgoing, where applicable).
Unit	governed reference data (minutes, MB, etc.).
Period	reporting interval, effective time semantics (Ch.5).
Revision	traffic observation reference, prior value, reason, actor, timestamp (Ch.25 Correction relationship).
Coverage State	expected-vs-received completeness per operator/period.

11.4 Relationships

Figure 7. Traffic Intelligence relationships to Revenue and Fusion.

SOURCE	RELATIONSHIP	TARGET
Traffic Observation	is an input to	Revenue Calculation (Ch.12)
Traffic Observation	is correlated by	Cross-Domain Fusion (Ch.19)
Revision	supersedes (with lineage)	prior Traffic Observation

11.5 Temporal and Provenance Semantics

A Revision never overwrites the original Traffic Observation; it creates a new version with a Correction relationship (Ch.25) so a Revenue calculation performed before the revision remains reproducible against the value it actually used, per Ch.5's bitemporal principle applied to a Revenue-relevant fact.

11.6 Failure / Exception Semantics

A missing period for an operator is modelled as Unknown coverage, never as zero traffic — this is load-bearing for Chapter 12's Revenue calculation, which must not silently compute a low Revenue figure from an absent Traffic Observation.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.11 in full.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm source granularity and frequency of traffic reporting per operator.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm units and direction conventions to standardise across operators.
- **[PROPOSED DESIGN DEFAULT]** Revisions are proposed to always be modelled as new versions with a Correction relationship (§11.5), never in-place overwrites; this is technically defensible from Ch.5/Ch.25 alone.

11.7 Acceptance Criteria

- A revised Traffic Observation is demonstrably retrievable in both its pre- and post-revision form.
- A missing period is demonstrably distinguishable from a zero-volume period in every consuming query.

Revenue Intelligence & Billing Verification Data Model

Defines revenue calculation inputs/results, international incoming traffic surcharge rules, billing verification and reconciliation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

12.1 Purpose

Defines revenue calculation inputs/results, the international incoming traffic surcharge rule, billing verification and reconciliation, implementing NCIE-002 Ch.12.

12.2 Semantic Definition

Calculated Revenue (derived deterministically by NCIE from Traffic and the effective-dated surcharge Rule) and Reported Revenue (supplied by the operator's Billing system) are distinct entities. A Reconciliation Difference between them is a labelled difference, never an automatic assertion of underpayment (NCIE-002 Ch.12 §12.4).

12.3 Principal Entities

Table 12. Revenue Intelligence & Billing Verification principal entities.

Entity	Attributes / Attribute Classes
Revenue Calculation	canonical ID, operator reference, period, traffic input reference(s) (Ch.11), surcharge Rule Version reference (Ch.24), calculated value, currency, calculation version.
Traffic Input Reference	revenue calculation reference, exact Traffic Observation version used (immutable pointer, per Ch.5 §5.6).
Surcharge Rule	See Ch.24 — this chapter references, and does not redefine, the Rule/Rule Version/Parameter entities.
Billing Verification	operator reference, period, reported value, currency, source reference.
Reconciliation	revenue calculation reference, billing verification reference, comparison basis.
Reconciliation Difference	reconciliation reference, difference value, status (candidate/explained/escalated), linked Case reference (Ch.16) if escalated.

12.4 The Surcharge Rate Is Not a Constant

The international incoming traffic surcharge — currently USD 0.19 (19 US cents) — is represented exclusively through the governed, effective-dated Rule / Rule Version / Parameter structure defined in Chapter 24. It is never stored as a literal constant in this chapter's entities, in application code, or in any UI configuration outside that governed structure. A Revenue Calculation references the specific Rule Version that was effective at its calculation period, so historical calculations remain correct and reproducible after the rate changes.

12.5 Relationships and Cardinality

- One Revenue Calculation references exactly one effective Surcharge Rule Version and one-to-many Traffic Input References.
- One Reconciliation compares exactly one Revenue Calculation with one Billing Verification for the same operator/period.
- A Reconciliation Difference may reference zero-or-one Case (Ch.16) if and when it is escalated to regulatory workflow.

12.6 Provenance and Temporal Semantics

Every Revenue Calculation is fully reproducible from its stored Traffic Input References and Rule Version reference alone (Ch.5, Ch.25) — recalculating with the same inputs must always yield the same result, which is the data-model precondition for NCIE-002 Ch.12's deterministic calculation requirement.

12.7 Failure / Exception Semantics

A Revenue Calculation cannot be produced for a period with Unknown Traffic coverage (Ch.11); it is marked not-calculable rather than computed from an implicit zero.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.12 in full, and depends on Ch.24's Rule/Rule Version/Parameter model for the surcharge rate.

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm the exact surcharge formula, legal basis and charging basis (per-minute, per-call, or other) with NCA Legal/Revenue Assurance.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm who may change the surcharge rate and who approves a rate change, to populate Ch.24's Rule approval workflow.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm billing-verification source system and the reconciliation workflow currently used, if any.

12.8 Acceptance Criteria

- A surcharge rate change is demonstrably effective-dated: Revenue Calculations before the change date reference the old Rule Version, after use the new one.
- A Reconciliation Difference is demonstrably surfaced with basis/unit and does not auto-populate a Case without explicit escalation.

Mobile Money Intelligence Data Model

Defines daily aggregate Mobile Money transfer amounts sent and received on each network.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

13.1 Purpose

Defines daily aggregate Mobile Money transfer amounts sent and received on each network, implementing NCIE-002 Ch.13. Subscriber-level data is out of scope, matching NCIE-001/002.

13.2 Semantic Definition

Sent and Received are modelled as separate measures on the same daily aggregate, never combined into a single net figure that would obscure their individual coverage/quality state.

13.3 Principal Entities

Table 13. Mobile Money Intelligence principal entities (deliberately minimal — no subscriber-level entity exists in this schema).

Entity	Attributes / Attribute Classes
Mobile Money Aggregate	canonical ID, operator/service reference, reporting date, sent amount, received amount, currency, source reference (Ch.6), coverage/revision state.
Operator/Service	reference to Operator (Ch.7) plus service identifier if the operator offers more than one Mobile Money service.

13.4 Schema-Level Exclusion of Subscriber Data

The Mobile Money Aggregate entity has no subscriber-identifying attribute, and no relationship path in this chapter connects it to any Protected Identity entity (Ch.14). This is a structural exclusion, not merely an access-control rule, so a future adapter cannot introduce subscriber-level data without a deliberate schema change to this chapter (implements NCIE-002 Ch.13 §13.4 at the data-model level).

13.5 Temporal Semantics

Mobile Money Aggregate carries reporting date, acquisition time and knowledge time (Ch.5); a missing day is Unknown coverage, never zero.

13.6 Failure / Exception Semantics

A revised aggregate for a prior date follows the same Correction relationship pattern as Chapter 11's Traffic Revision — the prior value remains retrievable.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.13 in full, including its schema-level subscriber-data exclusion principle.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm source system, currency and daily cut-off time for Mobile Money aggregate reporting.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm whether any additional approved aggregate measures (e.g. transaction counts) should be added to §13.3.

13.7 Acceptance Criteria

- A schema inspection confirms no subscriber-identifying attribute or relationship path exists anywhere in this chapter's entities.
- A missing day's aggregate is demonstrably distinguishable from a zero-volume day.

SIM Registration Intelligence Data Model

Defines SIM registrations, identity type, countable registrations, configurable limits and registration exceptions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

14.1 Purpose

Defines SIM registrations, identity type, countable registrations, configurable limits and registration exceptions, implementing NCIE-002 Ch.14.

14.2 Semantic Definition

- Ghana Card and Passport are distinct Identity Type values, never merged into one generic identity-document field, since their governed limits differ (§14.4).
- Raw identity values are Protected Identity References, structurally separate from the Registration count entity that uses them, so most queries never need to touch the protected reference at all.
- An identity exceeding its registration limit is a Registration Exception, not a fraud finding — culpability is never asserted by this entity's existence.

14.3 Principal Entities

Table 14. SIM Registration Intelligence principal entities.

Entity	Attributes / Attribute Classes
Registration	canonical ID, SIM reference, protected identity reference, operator reference (Ch.7), status, effective date.
SIM	canonical ID, operator reference, status.
Identity Type	governed enumeration (Ghana Card / Passport).
Protected Identity Reference	canonical ID (Ch.4), identity type, protected value (field-level access-controlled per NCIE-002 Ch.4/Ch.26).
Registration Status	governed enumeration (active/inactive/porting/deregistered).
Countable State	registration reference, whether this registration counts toward the identity's limit at a given date (some statuses may be excluded by governed rule).
Registration Rule	See Ch.24 — limit value, identity type, effective period; this chapter references, not redefines, it.
Registration Exception	protected identity reference, count at time of detection, applicable Registration Rule version, status (open/investigated/explained).

14.4 Limits Are Governed Rules, Not Constants

The Ghana Card maximum (currently 10) and foreign-passport maximum (currently 3) are represented exclusively through Chapter 24's Rule / Rule Version / Parameter structure, keyed by Identity Type, exactly as Chapter 12 requires for the surcharge rate. A Registration Exception references the specific Registration Rule version effective at its detection date, so a later rule change does not retroactively reclassify historical exceptions.

14.5 Lifecycle

Figure 8. Registration-to-exception detection lifecycle.

01	Registration recorded against Protected Identity Reference and SIM
02	Countable State evaluated against current Registration Rule version
03	Aggregate count per identity computed from Countable registrations

	▼
04	Count compared to effective Registration Rule limit
	▼
05	Registration Exception raised if count exceeds limit
	▼
06	Exception routed to Regulatory Intelligence (Ch.16) for investigation, not automatic action

14.6 Classification and Security

Protected Identity Reference is field-level access-controlled and never appears in aggregate reporting views; only the Registration count (not the underlying identity value) is exposed there, per NCIE-002 Ch.4 §4.3 and Ch.14 §14.5.

14.7 Failure / Exception Semantics

An ambiguous cross-operator identity match (Ch.4 §4.4 generalised) is never auto-merged before computing the aggregate count; it is retained as an unresolved candidate pending human confirmation, since a false merge could wrongly trigger or suppress a Registration Exception.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.14 in full, and depends on Ch.4 (identity resolution) and Ch.24 (governed limits).

Human Review Focus

- **[PROPOSED DESIGN DEFAULT]** The current supplied limits of 10 (Ghana Card) and 3 (Passport) are proposed as the initial Ch.24 Rule Service values; this is technically consistent with NCIE-001/002 and does not require new discovery to state as an initial default.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the exact lifecycle states counted toward the limit (§14.3 Countable State) — e.g. does a ported-out SIM still count?
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm field-level protection scope for Protected Identity Reference — which roles may view unmasked values, and under what legal authority.

14.8 Acceptance Criteria

- An identity exceeding the effective Registration Rule limit is demonstrably classified as a Registration Exception, not a fraud finding.
- A Registration Rule change is demonstrably effective-dated: exceptions detected before the change reference the prior rule version.

Anti-Fraud / SIMBOX Intelligence Data Model

Defines the data required to detect, investigate, locate and block SIMs associated with suspected SIMBOX activity.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

15.1 Purpose

Defines the data required to detect, investigate, locate and block SIMs associated with suspected SIMBOX activity, implementing NCIE-002 Ch.15's staged Anti-Fraud model.

15.2 Semantic Definition — Mandatory Separations

- Detection ≠ Confirmed Fraud — a SIMBOX Detection is a signal, never itself a fraud finding.
- Registered Identity ≠ Fraud Actor — the identity behind a Registration (Ch.14) is not asserted to be the operational fraud actor merely by association with a Suspected SIM.
- Blocking Requested ≠ Blocking Executed ≠ Blocking Verified — three distinct states, never collapsed into one 'blocked' flag.

15.3 Principal Entities

Table 15. Anti-Fraud / SIMBOX Intelligence principal entities.

Entity	Attributes / Attribute Classes
SIMBOX Detection	canonical ID, detection signal type, confidence, detected time, source reference (Ch.6).
Suspected SIM	SIM reference (Ch.14), linked detections, status.
Detection Evidence	evidence reference (Ch.17), linkage to SIMBOX Detection.
Investigation	canonical ID, assigned investigator (Ch.26), suspected SIM reference(s), status, sensitive-join log.
Tracking Authorization	investigation reference, authorizing actor, scope, effective period.
Location Evidence	investigation reference, geometry (Ch.8, heightened classification), source reference.
Blocking Request	investigation reference, target SIM, requested-by actor, status.
Blocking Action	blocking request reference, provider reference (Ch.23), execution status, timestamp.
Blocking Verification	blocking action reference, verification method, verified status.
Enforcement Outcome	investigation reference, outcome sourced only from the authoritative law-enforcement/regulatory system of record — never inferred by NCIE.

15.4 Stage Lifecycle

Figure 9. Anti-Fraud stage progression, each stage an explicit, auditable entity.

01	SIMBOX Detection (automated signal)
02	Suspected SIM (analyst-reviewed candidate)
03	Investigation opened (sensitive joins individually authorized and logged)
04	Location Evidence gathered under Tracking Authorization

05	Blocking Request raised, requiring separate validation and authorization
06	Blocking Action executed and Blocking Verification confirmed
07	Enforcement Outcome recorded only from the authoritative external system

15.5 Relationships and Cardinality

- One SIMBOX Detection may link to zero-or-more Suspected SIMs (a single signal can implicate multiple SIMs, e.g. a shared device).
- One Investigation has zero-or-more Blocking Requests, each with at most one Blocking Action and at most one Blocking Verification.
- Enforcement Outcome references the Investigation but is never derived by NCIE computation — only ingested from the authoritative external source.

15.6 Classification and Security

Location Evidence and Detection Evidence tied to Anti-Fraud investigations are sensitive-tier by default (NCIE-002 Ch.26). Sensitive joins (identity-to-location) require an explicit Tracking Authorization record before they may be performed, not merely a standing role grant.

15.7 Failure / Exception Semantics

A Blocking Action that fails at the provider is retried under idempotency control (Ch.28) and surfaces failure explicitly; it is never recorded as Verified without a genuine Blocking Verification.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.15 in full.

Human Review Focus

- [SOURCE/DATA DISCOVERY REQUIRED] Confirm actual Anti-Fraud workflow, tools and detection sources currently used at NCA.
- [SOURCE/DATA DISCOVERY REQUIRED] Confirm what location data is received and at what precision.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm blocking interfaces, approval matrix and verification process with operators.

15.8 Acceptance Criteria

- Each stage in §15.4 is demonstrably a distinct, independently queryable entity for a representative case.
- A Blocking Action cannot reach Verified status without both Blocking Request validation and a genuine Blocking Verification record.

Regulatory Case, Finding & Decision Data Model

Defines the formal regulatory objects through which intelligence becomes an institutional case, finding or decision.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

16.1 Purpose

Defines the formal regulatory objects through which intelligence becomes an institutional case, finding or decision, implementing NCIE-002 Ch.16.

16.2 Semantic Definition — Mandatory Separations

- Investigation/Situation ≠ Finding — an open Investigation (Ch.15) or Situation (Ch.18) does not automatically become a Finding merely by existing.
- Hypothesis ≠ Finding — a Hypothesis (Ch.20/Ch.21), including an ARGUS-attributed one, is working reasoning, never a Finding.
- Recommendation ≠ Decision — an ARGUS or analyst recommendation is advisory input to a Decision, never the Decision itself.
- Finding ≠ Decision — a Validated Finding states what is established; a Decision states the institutional action taken in response.

16.3 Principal Entities

Table 16. Regulatory Case, Finding & Decision principal entities.

Entity	Attributes / Attribute Classes
Case	canonical ID, originating domain/exception reference, status, assigned officer(s) (Ch.26).
Referral	case reference, referring entity, reason, timestamp.
Finding	case reference, state (proposed/validated), content, contributor origin (human/AI, Ch.20), validator (distinct from proposer, Ch.26 separation of duties).
Decision	case reference, finding reference(s), decision authority, content, status.
Decision Authority	role/actor reference (Ch.26), scope of authority.
Rationale	decision reference, reasoning content, evidence references (Ch.17).
Decision Status	governed enumeration (draft/issued/appealed/reversed).
Decision-Time Snapshot	decision reference, reconstructed state as of decision timestamp (Ch.5, Ch.25).
Review / Appeal	decision reference, prior decision reference (never overwritten), new decision reference, reason.

16.4 Case Lifecycle

Figure 10. Regulatory Case lifecycle from exception to decision.

01	Exception / Investigation / Situation (from any domain)
02	Case opened, Referral recorded if applicable
03	Proposed Finding drafted (AI contribution distinctly attributed per Ch.20)
04	Validated Finding (human validator distinct from proposer)

05	Decision issued by named Decision Authority, with Rationale and Decision-Time Snapshot
06	Appeal / Reversal preserves prior Decision and Finding, never overwrites

16.5 Relationships and Cardinality

- One Case has one-to-many Findings and zero-to-many Decisions over its lifecycle.
- A Review/Appeal always references its prior Decision; the prior Decision record is retained, never deleted, with the new Decision linked as superseding it.

16.6 Provenance and Security

A Finding’s contributor origin is preserved permanently — an AI-attributed contribution that a human adopts into a Validated Finding retains its AI-origin marker (NCIE-002 Ch.20 §20.6, Ch.23 §23.6). Decision Authority is checked against Ch.26’s role model at issuance time, not merely assumed from the acting user’s general role.

16.7 Failure / Exception Semantics

A Case with an existing Regulatory Intelligence platform outage (NCIE-002 Ch.16 §16.6) queues locally with Evidence intact rather than losing the Case; reconciliation on reconnection is recorded, not silently merged.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.16 in full, and depends on Ch.20 (AI-origin attribution) and Ch.26 (separation of duties).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm NCA case/finding terminology and existing Regulatory Intelligence platform’s own object model, to align field naming rather than duplicate it under different names.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm decision authorities and approval levels per case category.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm which decisions require preserved Rationale and Decision-Time Snapshot as a matter of regulatory defensibility.

16.8 Acceptance Criteria

- A Case is demonstrably blocked from reaching Validated Finding status without a distinct validator from the proposer.
- An appealed Decision demonstrably retains its full prior history intact.

Evidence & Evidence Relationship Model

Defines evidence objects, integrity, annotations, corrections, invalidation and relationships to reasoning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

17.1 Purpose

Defines evidence objects, integrity, annotations, corrections, invalidation and relationships to reasoning, implementing NCIE-002 Ch.23's Evidence Service.

17.2 Semantic Definition — Mandatory Separations

- Source Data ≠ Evidence — a raw Source Object Reference (Ch.6) becomes Evidence only when deliberately registered as such, not automatically upon ingestion.
- Annotation ≠ Evidence — a human or ARGUS annotation on an Evidence object is commentary, never itself independently authoritative Evidence.
- Invalidated Evidence is not necessarily deleted — invalidation is a state change, preserving the historical record of what was once relied upon.

17.3 Principal Entities

Table 17. Evidence & Evidence Relationship principal entities.

Entity	Attributes / Attribute Classes
Evidence Object	canonical ID, source reference (Ch.6), classification (Ch.31), status (active/invalidated).
Evidence Artifact	evidence object reference, content pointer, format.
Hash / Integrity Metadata	evidence artifact reference, hash algorithm, hash value, computed timestamp.
Annotation	evidence object reference, author (human/ARGUS, Ch.20), content, timestamp — explicitly not Evidence.
Evidence Relationship	source evidence reference, target reasoning-object reference (Hypothesis/Finding, Ch.16/Ch.20/Ch.21), relationship type (supports/contradicts).
Correction	evidence object reference, prior version reference, reason, actor, timestamp (Ch.5, Ch.25).
Invalidation	evidence object reference, reason, actor, timestamp — evidence object retained, status changed.
Classification	evidence object reference, tier (Ch.31).

17.4 Supporting and Contradictory Relationships

An Evidence Relationship of type contradicts must preserve both the supporting and contradicting Evidence's original sources — it never resolves the contradiction by discarding one side, consistent with NCIE-002 Ch.17's contradiction-preservation principle applied at the Evidence layer.

17.5 Lifecycle

Figure 11. Evidence lifecycle from registration to potential invalidation.

01	Source Object Reference deliberately registered as Evidence Object
02	Hash/Integrity Metadata computed at registration
03	Evidence Relationship links formed to Hypotheses/Findings as reasoning develops

	▼
04	Annotation added by human or ARGUS without altering the Evidence itself
	▼
05	Correction creates a new version if the Evidence Object requires amendment
	▼
06	Invalidation changes status without deleting the historical record

17.6 Provenance and Chain of Custody

Hash/Integrity Metadata is computed at registration and re-verified whenever the Evidence Artifact is retrieved for a formal proceeding, giving a verifiable chain of custody independent of the underlying storage technology (Ch.32).

17.7 Failure / Exception Semantics

A hash mismatch on retrieval is treated as a chain-of-custody failure requiring investigation, never silently ignored or auto-repaired.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.23 in full.

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] Confirm what NCA recognizes as formal Evidence for regulatory/legal purposes, to align the Evidence Object's classification and integrity requirements.
- [LEGAL/POLICY CONFIRMATION REQUIRED] Confirm chain-of-custody requirements in detail (algorithm, re-verification frequency).
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm retention and correction rules for Evidence, feeding Ch.31.

17.8 Acceptance Criteria

- An Annotation is demonstrably distinguishable from Evidence in every query and export.
- An Invalidation demonstrably preserves the original Evidence Object rather than deleting it.

REWS Alert, Situation & National Operating Picture Data Model

Defines Conditions, Alerts, Situations and the data required to compose the national situational picture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

18.1 Purpose

Defines Conditions, Alerts, Situations and the data required to compose the National Operating Picture, implementing NCIE-002 Ch.18.

18.2 Semantic Definition — Mandatory Separations

- Condition ≠ Alert ≠ Situation ≠ Investigation ≠ Case — five distinct entities, each with its own lifecycle (Ch.15, Ch.16 hold Investigation and Case; this chapter holds Condition, Alert and Situation).
- Acknowledgement is not resolution — acknowledging an Alert changes its acknowledgement state only, never its resolution status.
- National Operating Picture must preserve completeness, freshness and uncertainty as first-class properties, never implied by absence of data.

18.3 Principal Entities

Table 18. REWS Alert, Situation & National Operating Picture principal entities.

Entity	Attributes / Attribute Classes
Condition	canonical ID, domain reference, detected state, supporting evidence/observation references.
Alert	canonical ID, condition reference, alert rule reference (Ch.24), severity, raised time.
Severity	governed enumeration, effective-dated (Ch.24).
Acknowledgement	alert reference, actor, timestamp — distinct from resolution.
Escalation	alert reference, escalated-to actor/role, timestamp, reason.
Situation	canonical ID, grouped Alert/Incident/Evidence references, lifecycle status, owner.
Situation Association	situation reference, associated object reference (Alert/Incident/Evidence), relationship type.
NOP Projection	current composed view: per-domain known/missing/stale/waiting/contradictory status (Ch.2 vocabulary).
NOP Snapshot	NOP projection captured at a point in time, for historical reconstruction (Ch.5, Ch.25).

18.4 Lifecycle

Figure 12. Condition-to-Situation lifecycle feeding the National Operating Picture.

01	Observation/PM data evaluated against effective Alert Rule (Ch.24)
02	Condition detected
03	Alert raised with severity and evidence references
04	Acknowledgement recorded (distinct from resolution)
05	Escalation if unresolved within governed timing

06	Human groups related Alerts/Incidents/Evidence into a Situation
07	NOP Projection recomputed and periodically captured as an NOP Snapshot

18.5 Relationships and Cardinality

- One Condition may produce zero-or-one Alert per effective Alert Rule (avoiding duplicate alerts for the same underlying condition).
- One Situation has one-to-many Situation Associations, spanning Alerts, Incidents (Ch.10) and Evidence (Ch.17) across multiple domains.

18.6 Provenance and Temporal Semantics

Every Alert retains its triggering Alert Rule version (Ch.24) so historical Alerts remain explicable even after rules change. NOP Snapshots allow reconstruction of "what the national picture looked like" at a prior time, supporting Ch.25's decision-time reconstruction needs at the situational-awareness level.

18.7 Failure / Exception Semantics

Loss of the REWS evaluation pipeline is reflected as a stale/waiting NOP Projection state, never as a silently frozen 'all clear' picture (NCIE-002 Ch.18 §18.8).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.18 in full, and depends on Ch.24 for governed Alert Rules.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm REWS severity taxonomy and escalation timing per category.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm Alert acknowledgement/escalation workflow and closure authority.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm which domains are mandatory for a complete National Operating Picture at launch.

18.8 Acceptance Criteria

- An Alert is demonstrably traceable to the Condition and evidence that triggered it.
- An NOP Snapshot is demonstrably reconstructable for a prior point in time, including its uncertainty state at that time.

Cross-Domain Fusion & Intelligence Graph Model

Defines how objects from different domains may be connected for cross-domain analysis without losing their native meaning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

19.1 Purpose

Defines how objects from different domains may be connected for cross-domain analysis without losing their native meaning, implementing NCIE-002 Ch.17.

19.2 Semantic Definition — Mandatory Separation

Graph Relationship ≠ Causality. Every edge in the intelligence graph states a correlation type and its supporting basis; none is rendered or queryable in a way that implies a causal claim the evidence does not support.

19.3 Principal Entities

Table 19. Cross-Domain Fusion & Intelligence Graph principal entities.

Entity	Attributes / Attribute Classes
Graph Node Reference	pointer to a canonical entity in its owning domain chapter — the graph never copies the entity's substantive attributes.
Typed Edge	source node reference, target node reference, relationship type.
Edge Origin	governed enumeration: observed / deterministic / human-inferred / ARGUS-proposed — each requiring a different validation state per §19.4.
Validation State	edge reference, state (unvalidated/human-confirmed/rejected).
Temporal Validity	edge reference, effective period during which the relationship held (Ch.5).
Evidence Support	edge reference, evidence object reference(s) (Ch.17).
Cross-Domain Correlation	a computed set of edges meeting a correlation pattern, with confidence.
Graph Projection	a bounded, queryable view of the graph for a given scope/time window.

19.4 Edge Origin and Validation

Table 20. Required validation state by edge origin before an edge may support a Finding.

Edge Origin	Required Validation State Before Use in a Finding
Observed	Directly evidenced; usable as supporting Evidence Support without further validation.
Deterministic	Computed by a governed, deterministic rule (Ch.24); usable once the rule itself is approved.
Human-Inferred	Requires the inferring human's attribution retained; usable as working analysis, requires promotion (Ch.16) to support a Finding.
ARGUS-Proposed	Requires explicit AI-origin attribution (Ch.20) and human validation before any use beyond working analysis; never auto-promoted.

19.5 Relationships and Cardinality

A Typed Edge references exactly two Graph Node References (source, target) and zero-to-many Evidence Support records. Multiple edges of different types may exist between the same two nodes without merging into one relationship.

19.6 Provenance

Every edge retains its Edge Origin permanently, even after human validation — validating an ARGUS-Proposed edge does not silently reclassify it as Observed or Deterministic (mirrors the AI-origin-preservation principle of Ch.16 §16.6 and Ch.20).

19.7 Failure / Exception Semantics

Two edges that conflict (e.g. contradictory correlation claims about the same two nodes) are both retained with their respective Evidence Support, never auto-resolved.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.17 in full, and depends on Ch.20 for ARGUS-Proposed edge attribution.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which relationship types are operationally useful for launch (§19.3 Typed Edge).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which inferred edges require mandatory human validation before appearing in any shared view.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm sensitive graph traversal rules where a path would cross a protected-identity or Anti-Fraud boundary (Ch.14, Ch.15).

19.8 Acceptance Criteria

- An ARGUS-Proposed edge is demonstrably blocked from supporting a Finding until explicitly human-validated.
- A sampled Graph Projection demonstrably never displays a causal claim unsupported by its Edge Origin and Evidence Support.

ARGUS Interaction & AI Provenance Data Model

Defines data needed to govern ARGUS participation, tools, model routing and AI-originated analytical objects.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

20.1 Purpose

Defines data needed to govern ARGUS participation, tools, model routing and AI-originated analytical objects, implementing NCIE-002 Ch.19 (Intelligence Assistant), Ch.19 §19.11 (ARGUS Naming Boundary) and Ch.24 (AI Platform).

20.2 Semantic Definition — Mandatory Separations

- ARGUS output is not automatically Evidence (Ch.17) — an ARGUS Contribution becomes Evidence only if it is itself registered as such through deliberate promotion, which is rare and distinct from ordinary use.
- AI origin must remain after human adoption or modification — a Human Modification of an AI Contribution retains the original AI Contribution's identity and content, never overwriting it.
- Private model chain-of-thought is not persisted. This chapter's entities capture defensible reasoning artifacts and provenance — sources, tools, outputs and governance metadata — never the model's internal, unexaminable deliberation (NCIE-002 Ch.23 §23.6, Ch.24 §24.4).

20.3 Principal Entities

Table 21. ARGUS Interaction & AI Provenance principal entities (all mandated classes from the handover instructions are represented).

Entity	Attributes / Attribute Classes
ARGUS Interaction	canonical ID, conversation/room reference, participant (human or ARGUS), timestamp.
ARGUS Contribution	interaction reference, contribution type, content, AI Origin marker.
Contribution Type	governed enumeration: Question / Challenge / Hypothesis / Contradiction / Evidence Gap / other analytical contribution types fixed by NCIE-002 Ch.20 §20.3.
Tool Invocation	contribution reference, tool name, parameters, authorization check result (Ch.26), response reference.
Model Route / Version	interaction reference, model/provider reference (Ch.24 Model Registry), version, retained where governance requires audit.
Evidence Reference	contribution reference, evidence object reference(s) (Ch.17) — a contribution cites Evidence; it does not become Evidence.
Human Response	contribution reference, responding human, action (accept/reject/modify/defer/promote).
Human Modification of AI Contribution	original contribution reference (retained, unmodified), modifying human, modified content, timestamp.
Participation Mode	room/interaction reference, mode (Participatory/Invoked/Evidence-Watch/Silent, Ch.21 §21.7), effective period.
AI Provenance	contribution reference, full derivation chain: sources cited, tools invoked, model route, human responses — never internal model reasoning.

20.4 Human Modification Never Erases AI Origin

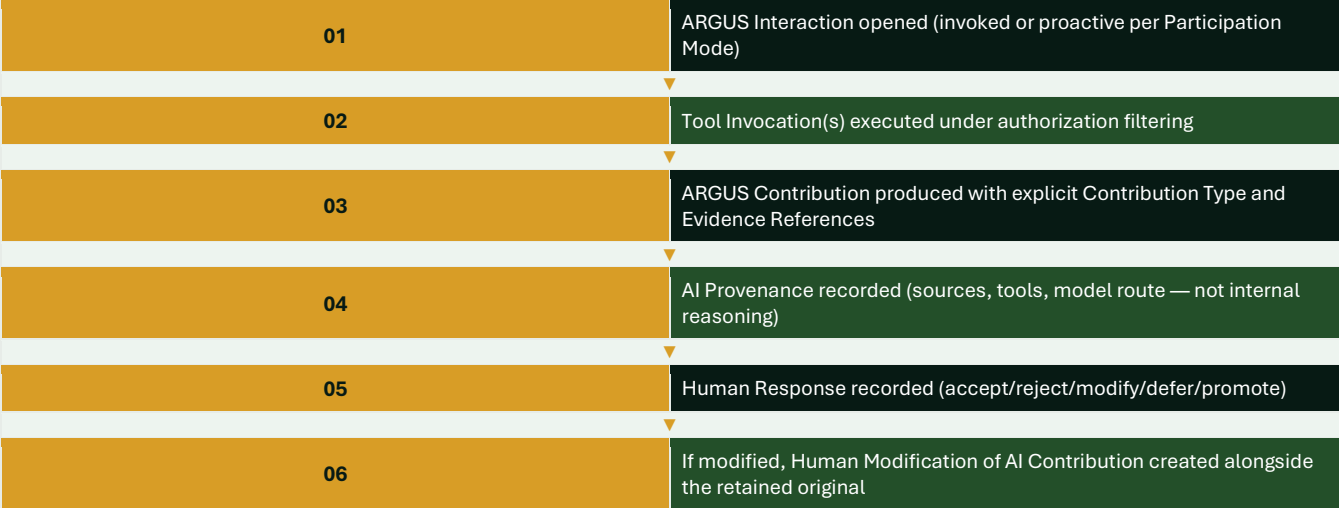
When a human edits or extends an ARGUS Contribution, the system creates a Human Modification of AI Contribution record that references, and does not replace, the original ARGUS Contribution. Any later display of the modified content shows both the current text and its AI-originated ancestry, directly implementing NCIE-002 Ch.20 §20.6 and Ch.23 §23.6.

20.5 Tool Authorization at the Data Layer

Every Tool Invocation stores its authorization check result at the time of invocation, not only a boolean success flag — this gives Ch.27's audit model a queryable record of exactly which authorization layer (Ch.26) permitted or denied each ARGUS tool use, implementing NCIE-002 Ch.19 §19.7 and Ch.24 §24.4's guardrail requirements.

20.6 Lifecycle

Figure 13. ARGUS interaction-to-provenance lifecycle.



20.7 Classification and Security

An ARGUS Contribution inherits the classification of the Evidence and context it references; it is never rendered at a lower classification than its most sensitive input, consistent with NCIE-002 Ch.19 §19.7's authorization-filtering requirement.

20.8 Failure / Exception Semantics

If the Model Gateway is unavailable mid-interaction, the ARGUS Interaction record closes with an explicit degraded-completion status rather than a silently truncated or fabricated continuation (NCIE-002 Ch.19 §19.8).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.19 (Assistant), Ch.19 §19.11 (naming boundary), Ch.20 (collaborative reasoning) and Ch.24 (AI Platform, Model Gateway, Guardrails).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm retention period for ARGUS Interactions and Contributions.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which model/tool metadata (§20.3 Model Route/Version) NCA requires retained for audit versus what can be summarised.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm treatment of private versus shared ARGUS interactions under applicable records/privacy policy.

20.9 Acceptance Criteria

- A sampled ARGUS Contribution demonstrably carries Contribution Type, AI Origin marker and Evidence References, with no persisted private chain-of-thought field anywhere in the schema.
- A human-modified ARGUS Contribution demonstrably retains its original, unmodified AI-originated version alongside the modification.

Collaborative Intelligence & Multi-User Brainstorming Data Model

Defines persistent collaborative Rooms in which humans and ARGUS can brainstorm, challenge, hypothesize and preserve structured reasoning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

21.1 Purpose

Defines persistent collaborative Rooms in which humans and ARGUS can brainstorm, challenge, hypothesize and preserve structured reasoning, implementing NCIE-002 Ch.20 in full. This chapter is deliberately not reducible to chat messages: it defines first-class persistent objects so collaborative intelligence can stop, resume and evolve without depending entirely on transcript history.

21.2 Semantic Definition — Mandatory Separations

- Room is more than chat — a Room's Collaborative Reasoning State (§21.4) is a structured object distinct from its raw message transcript.
- Shared disclosure must not become permission union — a Room's shared context is filtered per-participant by that participant's own authorization (Ch.26), never the union of all participants' permissions.
- Consensus ≠ Finding — participants agreeing within a Room produces a Consensus record, which remains working analysis until deliberately promoted through Ch.16's Finding Workflow.

21.3 Principal Entities (Mandatory Persistent Object Classes)

Table 22. Collaborative Intelligence & Multi-User Brainstorming principal entities — every class listed in the authoritative handover instructions is represented as a first-class persistent object.

Entity	Attributes / Attribute Classes
Collaborative Room	canonical ID, room type, classification (Ch.31), current Participation Mode (Ch.20 §20.5, Ch.21 §21.7).
Participant	room reference, user reference (Ch.26), individual permission scope, join/leave timestamps.
Thread	room reference, topic, status — supports multiple concurrent analytical threads without cross-contamination.
Question	thread reference, originator (human or ARGUS via Ch.20), content.
Hypothesis	thread reference, originator, content, supporting/contradicting Evidence References (Ch.17), status (active/adopted/deferred/rejected).
Assumption	thread reference, originator, content, whether tested.
Contradiction	thread reference, conflicting Hypothesis or Evidence references, status.
Evidence Gap	thread reference, description of missing evidence, originator.
Evidence Relationship	thread reference, evidence object reference (Ch.17), relationship to Hypothesis/Assumption (supports/contradicts) — shares its type vocabulary with Ch.17 §17.3.
Task	thread reference, description, assignee, status.
Waiting Dependency	thread reference, description (e.g. WAITING_FOR_PM), linked Workflow reference (Ch.23), status.
Human Judgment	thread reference, judging human, content — a human's own analytical position, distinct from a Hypothesis attributed to ARGUS.
Decision Point	thread reference, description of a choice the room must make, linked options.
ARGUS Intervention	thread reference, ARGUS Contribution reference (Ch.20), triggering condition, Participation Mode in effect at the time.

Entity	Attributes / Attribute Classes
Consensus	thread reference, participants agreeing, content — explicitly not a Finding (§21.2).
Dissent	thread reference, dissenting participant, content — preserved even where a Consensus exists.
Room Snapshot	room reference, timestamp, captured Collaborative Reasoning State — supports 'where were we?' reconstruction (Ch.5, Ch.25).

21.4 Collaborative Reasoning State

A Room's Collaborative Reasoning State is the composed, queryable aggregation of its current Threads, Questions, Hypotheses, Assumptions, Contradictions, Evidence Gaps, Tasks, Waiting Dependencies, Human Judgments, Decision Points, ARGUS Interventions, Consensus and Dissent records. It is distinct from — and reconstructable independently of — the raw message transcript, directly implementing NCIE-002 Ch.20 §20.5.

21.5 Multi-Speaker Attribution

Every object in §21.3 that has an originator field preserves that originator permanently, including explicit AI origin where the originator is ARGUS (via Ch.20's AI Origin marker). Adoption, challenge or modification by another participant does not overwrite the original originator field (NCIE-002 Ch.20 §20.6).

21.6 Analytical Thread Isolation

Every object in §21.3 references exactly one Thread. Evidence, Hypotheses and Contradictions belonging to different Threads within the same Room are never implicitly combined in a query unless the query explicitly spans threads, preventing cross-thread contamination (NCIE-002 Ch.20 §20.6).

21.7 Participation Modes and Room-to-Workflow Linkage

Participation Mode (Participatory/Invoked/Evidence-Watch/Silent) is stored per Room and changeable by an authorized participant, per NCIE-002 Ch.20 §20.7. A Waiting Dependency links to its originating Workflow (Ch.23) via the Room-to-Workflow Bridge pattern, so a validated result can return asynchronously to the correct Thread without re-establishing context (NCIE-002 Ch.21 §21.4).

21.8 Permission-Aware Shared Context

The Room's effective shared disclosure context is computed per-participant at query time from each Participant's own permission scope (Ch.26) — it is never stored as a single pooled permission set for the Room, implementing NCIE-002 Ch.20 §20.9 at the data-model level.

21.9 Room Snapshot and Cross-Session Continuity

A Room Snapshot captures the Collaborative Reasoning State at a point in time (e.g. end of shift), enabling 'where were we?' reconstruction by combining the snapshot with current authoritative NCIE state, per NCIE-002 Ch.20 §20.10 — never presenting a stale snapshot as current without reconciliation.

21.10 Failure / Exception Semantics

Concurrent edits to the same Thread's Collaborative Reasoning State are resolved with a visible edit trail (last-writer-wins per field plus history), never silent data loss, matching NCIE-002 Ch.20 §20.16.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.20 in full, and depends on Ch.20 of this document (ARGUS Interaction) for AI-originated object attribution.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm Room types and membership rules (who may create a Room, invite participants).

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm default Participation Mode per Room type (ordinary vs Anti-Fraud/regulatory rooms, per NCIE-002 Ch.20's proposed default).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm how Consensus, Dissent and private-to-room sharing should work in edge cases (e.g. a participant leaving mid-discussion).

21.11 Acceptance Criteria

- A scripted two-human, mixed-permission Room session demonstrates permission-aware filtering (§21.8), correct multi-speaker attribution (§21.5) and thread isolation (§21.6).
- A Room Snapshot is demonstrably sufficient, combined with current authoritative state, to answer 'where were we?' without relying on the raw transcript.

Memory & Institutional Knowledge Data Model

Defines long-term memory classes and how reviewed experience becomes reusable Institutional Knowledge.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

22.1 Purpose

Defines long-term memory classes and how reviewed experience becomes reusable Institutional Knowledge, implementing NCIE-002 Ch.20 §20.2 (memory tiers) and Ch.20 §20.12 (formal promotion boundary).

22.2 Semantic Definition — Mandatory Separations

- Memory is not current truth — a Memory Object records what was previously observed, decided or discussed; it is never queried as if it were live authoritative state.
- ARGUS is not institutional memory — ARGUS has no persistent memory store of its own separate from the governed Memory Object entities defined here; anything ARGUS 'recalls' is retrieved from these entities under normal authorization.
- Institutional Knowledge ≠ Rule — a piece of Institutional Knowledge (e.g. "district X historically degrades during rainy season") informs human and ARGUS reasoning but is never itself a governed Rule (Ch.24) unless deliberately encoded as one.

22.3 Principal Entities

Table 23. Memory & Institutional Knowledge principal entities.

Entity	Attributes / Attribute Classes
Memory Object	canonical ID, memory class, owner (user/room/institutional), content reference, recorded time.
Memory Class	governed enumeration: Private User Memory / Shared Room Memory / Institutional Memory (NCIE-002 Ch.20 §20.2).
Knowledge Candidate	memory object reference, proposed generalisation, proposing actor — requires human promotion (§22.4).
Institutional Knowledge	knowledge candidate reference (once promoted), content, applicability scope, review state.
Knowledge Version	institutional knowledge reference, version, change reason (Ch.5, Ch.25 pattern).
Applicability Scope	institutional knowledge reference, domain(s)/geography/time window the knowledge applies to.
Review State	institutional knowledge reference, reviewer, decision, timestamp.
Supersession / Retraction	institutional knowledge reference, prior version reference, reason — history remains traceable.

22.4 Promotion Requires Human Action

A Knowledge Candidate — whether proposed by a human or by ARGUS (Ch.20) — never becomes Institutional Knowledge automatically. Promotion is a deliberate, audited human action recorded as a Review State entry, implementing NCIE-002 Ch.20 §20.12's formal promotion boundary.

22.5 Lifecycle

Figure 14. Memory-to-Institutional-Knowledge promotion lifecycle.

01	Memory Object recorded (private/shared/institutional per its Memory Class)
----	--

02	Knowledge Candidate proposed from a pattern noticed in memory or collaborative reasoning (Ch.21)
03	Human review (Review State)
04	Promotion to Institutional Knowledge, or rejection/deferral
05	Superseded/retracted Institutional Knowledge remains historically traceable via Knowledge Version

22.6 Sharing and Classification

Private User Memory is never automatically shared into a Room merely because it would be relevant, matching Ch.21 §21.8's permission-aware sharing rule. Institutional Knowledge is classified and access-controlled exactly as its Applicability Scope's domain requires (Ch.26, Ch.31) — e.g. Institutional Knowledge derived from Anti-Fraud experience remains sensitive-tier.

22.7 Failure / Exception Semantics

Superseded or retracted Institutional Knowledge is never deleted; it is marked superseded/retracted with a reason and remains queryable for historical reconstruction of what was believed true at a past point in time (Ch.5, Ch.25).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.20 §20.2 and §20.12 in full.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm memory retention period by Memory Class.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm Institutional Knowledge owners, reviewers and approvers.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Confirm what may become organisation-wide Institutional Knowledge versus remaining room- or user-scoped.

22.8 Acceptance Criteria

- A Knowledge Candidate is demonstrably blocked from becoming Institutional Knowledge without a recorded human Review State.
- A superseded piece of Institutional Knowledge is demonstrably retrievable in its pre-supersession form.

Workflow, Automation, RPA & Tasking Data Model

Defines persistent workflows, approvals, tasks, waiting dependencies and automated execution state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

23.1 Purpose

Defines persistent workflows, approvals, tasks, waiting dependencies and automated execution state, implementing NCIE-002 Ch.21's Orchestrator, Scheduler and Workflow Engine.

23.2 Semantic Definition — Mandatory Separations

- Requested ≠ Approved ≠ Executed ≠ Verified — four distinct states on the same Workflow Instance, never collapsed into a single 'done' flag.
- Unknown external outcome requires reconciliation rather than blind retry — if an automated action's outcome cannot be confirmed, the system records Unknown and schedules reconciliation, never silently re-executing an action that may have already succeeded.

23.3 Principal Entities

Table 24. Workflow, Automation, RPA & Tasking principal entities.

Entity	Attributes / Attribute Classes
Workflow Instance	canonical ID, workflow definition reference, originator (human/room/schedule, Ch.21 of this document), current state.
Workflow State	governed enumeration: Requested / Approved / Executed / Verified / Failed / Reconciling.
Task	workflow instance reference, description, assignee, status — includes Human Tasks (NCIE-002 Ch.21).
Approval	workflow instance reference, approving actor, decision, timestamp — separate from the requesting actor (Ch.26 separation of duties).
Waiting Dependency	workflow instance reference, named wait state (e.g. WAITING_FOR_PM, WAITING_FOR_TRAFFIC), entered/resolved timestamps.
Automation/RPA Run	workflow instance reference, provider reference (Ch.6-style provider abstraction), bot identity (Ch.26 machine identity), idempotency key, output artifact reference.
External Reference	run reference, external system's own reference/ticket ID for correlation.
Outcome	run reference, result (success/failure/unknown), evidence reference if applicable (Ch.17).
Reconciliation State	run reference, reconciliation status, reconciled-by actor/process, timestamp.

23.4 State Machine

Figure 15. Workflow state machine from request to verified outcome.

01	Workflow Instance Requested (human, room-suggested per Ch.21, or scheduled)
02	Approval recorded by a distinct authorizing actor where required
03	Automation/RPA Run executed with idempotency key
04	Outcome recorded (success/failure/unknown)

	▼
05	If Unknown, Reconciliation State entered rather than automatic retry
	▼
06	Workflow Instance reaches Verified only once Outcome is confirmed

23.5 Room-Originated Workflows

A Workflow Instance originated by a Collaborative Room (Ch.21 of this document) retains its originating Room/Thread reference for provenance, and its Waiting Dependency may be observed by that Room in Evidence-Watch Participation Mode without granting the Room any execution authority — the AI-suggested-is-not-execution-authority principle is enforced by requiring a distinct Approval actor, never by the room-suggestion alone (NCIE-002 Ch.20 §20.11, Ch.21 §21.4).

23.6 Idempotency and Correlation

Every Automation/RPA Run carries an idempotency key so a duplicated trigger (e.g. a re-delivered event, Ch.28) cannot cause the same external action to execute twice. External Reference correlates NCIE's Run with the external system's own ticket/reference for reconciliation.

23.7 Failure / Exception Semantics

A provider outage mid-Run leaves the Workflow Instance in a persisted Waiting Dependency or Reconciling state that survives a platform restart (NCIE-002 Ch.28) — it is never lost or silently marked Failed without an attempted reconciliation first.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.21 in full, and depends on Ch.21 of this document (Room-to-Workflow linkage) and Ch.26 (approval separation of duties).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which workflows are high-consequence and require mandatory Approval before Automation/RPA Run execution.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm approval matrices per workflow category.
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm which external systems require RPA versus API integration today.

23.8 Acceptance Criteria

- A Workflow Instance is demonstrably blocked from reaching Executed without a recorded Approval where one is required.
- A simulated duplicate trigger is demonstrably prevented from producing a duplicate Automation/RPA Run via the idempotency key.

Rules, Configuration & Calculation Data Model

Defines governed rules and configurable parameters used by deterministic NCIE calculations and thresholds.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

24.1 Purpose

Defines governed rules and configurable parameters used by deterministic NCIE calculations and thresholds, implementing NCIE-002 Ch.22. This chapter is the single authoritative definition referenced — not redefined — by Chapters 7, 12, 14 and 18.

24.2 Semantic Definition — Mandatory Principle

Rates and limits belong in versioned Rules, never in UI constants or application code. This applies without exception to the international incoming traffic surcharge (currently USD 0.19), the Ghana Card registration limit (currently 10) and the foreign-passport registration limit (currently 3): each is a Parameter value inside a governed Rule Version, not a literal anywhere else in the system.

24.3 Principal Entities

Table 25. Rules, Configuration & Calculation principal entities.

Entity	Attributes / Attribute Classes
Rule	canonical ID, rule key (e.g. international-surcharge, sim-limit-ghana-card, sim-limit-passport, kpi-threshold-*, rews-alert-rule-*), domain.
Rule Version	rule reference, value/Parameter set, effective-from/effective-to (Ch.5), approval state.
Parameter	rule version reference, name, value, unit/currency.
Effective Period	rule version reference, start date, end date (nullable for open-ended).
Approval State	rule version reference, approving actor, timestamp.
Calculation Formula Reference	links a Rule (e.g. KPI or Revenue formula) to the calculation logic version it governs (Ch.33 schema versioning applies to the formula's own contract).
Scenario Rule	a what-if overlay Rule Version, explicitly isolated from production Rule Version history (§24.5).
Rule Conflict State	rule reference, conflicting rule versions detected, resolution status — blocks authoritative output while unresolved.

24.4 Effective Dating and Historical Binding

A calculation (Revenue, Ch.12; KPI evaluation, Ch.7; Registration Exception detection, Ch.14; Alert Rule evaluation, Ch.18) always binds to the Rule Version effective at its own reference date, and stores that Rule Version reference permanently — a later rule change never retroactively alters a historical calculation's stored result.

24.5 Scenario Isolation

A Scenario Rule exists only within a Scenario Time context (Ch.5 §5.2) and is never written to, or read as, production Rule Version history. A what-if Revenue projection using a Scenario Rule is clearly labelled as a Scenario result and cannot be mistaken for an authoritative Revenue Calculation (Ch.12).

24.6 Conflict Handling

Two Rule Versions with overlapping Effective Periods for the same rule key are a Rule Conflict State: authoritative output for the affected calculation is blocked, not silently resolved by picking one, until an administrator resolves the conflict (NCIE-002 Ch.22 §22.4).

24.7 Relationships and Cardinality

- One Rule has one-to-many Rule Versions over time, at most one of which is effective at any given non-conflicted point in time.
- One Rule Version has one-to-many Parameters (e.g. the SIM-limit rule's Parameter set might include both the numeric limit and its identity-type key).

24.8 Failure / Exception Semantics

If the Rule Service is unavailable, dependent Calculation Engines use the last successfully cached Rule Version with a visible staleness indicator — they never fall back to a hardcoded default value (NCIE-002 Ch.22 §22.8).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.22 in full, and is depended upon by Ch.7, Ch.12, Ch.14 and Ch.18 of this document.

Human Review Focus

- **[PROPOSED DESIGN DEFAULT]** The current supplied values (surcharge USD 0.19; Ghana Card limit 10; Passport limit 3) are proposed as the initial Rule Version values at go-live, per NCIE-002's existing baseline.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which values users may modify at what administrative level, and the approval/effective-date process for each rule key.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm rule-conflict and rollback behaviour expectations beyond the block-on-conflict default in §24.6.

24.9 Acceptance Criteria

- A rate/limit change is demonstrably effective-dated: calculations before the change date use the prior Rule Version, after use the new one, with both retrievable.
- An overlapping-effective-date Rule Conflict State is demonstrably blocked from producing authoritative output.

Provenance, Lineage & Decision-Time Data Model

Defines end-to-end source lineage and dependency structures needed for defensible historical reconstruction.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

25.1 Purpose

Defines end-to-end source lineage and dependency structures needed for defensible historical reconstruction, implementing NCIE-002 Ch.23's Provenance Graph and Historical Snapshot Service, and providing the Correction/Lineage mechanism referenced by Chapters 5, 6, 11, 12 and 14.

25.2 Semantic Definition

Lineage supports both backward tracing ("what did this fact derive from?") and forward impact analysis ("what depends on this fact, and what breaks if it is corrected?"). Decision-Time (Ch.5, Ch.16) must exclude later Evidence by construction, not by query-time filtering alone.

25.3 Principal Entities

Table 26. Provenance, Lineage & Decision-Time principal entities.

Entity	Attributes / Attribute Classes
Lineage Record	canonical ID, from-object reference, to-object reference, transformation type, actor (human/AI/bot/service, Ch.20/Ch.26).
Transformation Record	lineage record reference, transformation logic/version reference (Ch.33).
Source Reference	lineage record reference, originating Source Object Reference (Ch.6).
Dependency Edge	object reference, depends-on object reference — supports forward impact analysis.
Snapshot	reference date/time, reconstructed state (Ch.5's Historical Snapshot Service data).
Decision-Time Context	decision reference (Ch.16), bounded fact/evidence/rule-version set available at the decision's timestamp.
Correction Relationship	corrected object reference, prior version reference, reason, actor, timestamp — the mechanism Chapters 5/11/12/14 each reference.
Impact-Review State	dependency edge reference, review status after an upstream correction — flags downstream objects for human review, never silently recalculates them.

25.4 Forward Impact Analysis on Correction

When a Correction Relationship is created for an upstream fact (e.g. a Traffic Observation Revision, Ch.11), the Dependency Edges pointing to it are walked to identify downstream objects (e.g. a Revenue Calculation, Ch.12) whose Impact-Review State is then set to flagged — those downstream objects are never silently recalculated without human review, since a silent recalculation could alter a figure already relied upon in a Decision (Ch.16).

25.5 Decision-Time Construction

A Decision-Time Context is built at the moment a Decision (Ch.16) is issued, by binding to the then-current Rule Versions (Ch.24), Evidence (Ch.17) and canonical facts with a knowledge time at or before the Decision's timestamp — later Evidence is structurally excluded from the bound set, not merely filtered out at display time, so it cannot leak in through a different query path.

25.6 Relationships and Cardinality

One object may have many Lineage Records leading to it (multiple sources/transformations contributing to one canonical fact) and many Dependency Edges leading from it (many downstream consumers). A Snapshot references the full set of objects and their state as of its timestamp, not a partial subset.

25.7 Failure / Exception Semantics

A Lineage Record that cannot be constructed because the originating Source Object Reference was purged (per an approved retention rule, Ch.31) retains a tombstone reference recording that purge, rather than leaving a silent gap that looks like a modelling defect.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.23 in full (lineage, decision-time reconstruction, historical snapshots).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm how much lineage is mandatory by object class (full transformation chain for Revenue and Findings; lighter for high-volume raw telemetry).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm archival period for Snapshots, feeding Ch.31/Ch.35.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which Decisions require formal Decision-Time Context reconstruction versus a lighter-weight audit trail.

25.8 Acceptance Criteria

- A correction to an upstream fact demonstrably flags its downstream dependents for review rather than silently recalculating them.
- A Decision-Time Context reconstruction demonstrably excludes Evidence with a knowledge time after the Decision's timestamp.

Identity, Authorization & Security Data Model

Defines workforce/service identities, roles, attributes, permissions, delegated authority and data classification relationships.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

26.1 Purpose

Defines workforce/service identities, roles, attributes, permissions, delegated authority and data classification relationships, implementing NCIE-002 Ch.4 (Identity, Authorization & Zero-Trust) at the data-model level, and providing the User/Role/Machine Identity entities referenced throughout this document.

26.2 Semantic Definition — Mandatory Separations

- Room membership does not automatically grant Evidence access — a Participant's Room membership (Ch.21) and their domain-level Object-Access Relation to a specific Evidence Object (Ch.17) are separate grants.
- Current permission is not historical permission — a permission change is effective-dated (Ch.5), so a past authorization decision remains explicable using the permission state that applied at that time, not today's.
- Human, service, workflow and ARGUS actors remain distinguishable in every access/action record — no actor-identity field is ever generic enough to blur these four classes.

26.3 Principal Entities

Table 27. Identity, Authorization & Security principal entities.

Entity	Attributes / Attribute Classes
User	canonical ID, human identity reference, status.
Service Identity	canonical ID, service/workflow/ARGUS machine identity reference, distinct credential lifecycle from Users (NCIE-002 Ch.4 §4.2).
Role	canonical ID, name, bundle of domain/object/field/action grants, versioned.
Permission	role reference, domain, object type, field-level scope, action(s) permitted.
Attribute	user/service reference, attribute name/value used in contextual authorization (e.g. clearance level).
Delegation	delegating actor, delegate actor, scope, effective period.
Classification	See Ch.31 — this chapter references classification tiers rather than redefining them.
Object-Access Relation	actor reference, object reference, effective period, granting basis (role/delegation/exception).
Authentication / Session Reference	user/service reference, session token reference, MFA status, issued/expiry time.

26.4 Effective-Dated Permissions

A Permission granted through a Role, and any Delegation, is effective-dated exactly as Chapter 24's Rules are — an audit or Decision-Time reconstruction (Ch.25) queries the Object-Access Relation as it stood at the relevant historical timestamp, not the current Role definition.

26.5 Four Distinguishable Actor Classes

Table 28. The four actor classes every access/action record must distinguish.

Actor Class	Represented By	Never Confused With
Human	User entity, authenticated Session Reference.	Never inherits a Service Identity's credential.

Actor Class	Represented By	Never Confused With
Service	Service Identity, scoped machine credential (NCIE-002 Ch.4 §4.2).	Never assumes a User's session.
Workflow	Service Identity scoped to a Workflow Instance (Ch.23).	Distinguishable from ARGUS even where both act 'on behalf of' the same Workflow.
ARGUS	Service Identity per NCIE-002 Ch.19 §19.11's stable internal identity, independent of its display name.	Never recorded merely as 'system' or merged with generic Service Identity records.

26.6 Relationships and Cardinality

A User has one-to-many Roles; a Role has one-to-many Permissions. An Object-Access Relation references exactly one actor (User or Service Identity) and one object, with its granting basis recorded so a later audit can tell whether access came from a standing Role or a time-boxed Delegation.

26.7 Failure / Exception Semantics

A Delegation whose effective period has lapsed is never silently treated as still active; Object-Access Relation evaluation checks the Delegation's Effective Period at query time (NCIE-002 Ch.4 §4.4).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.4 in full, and is depended upon by every chapter of this document that references role, permission or actor identity.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm enterprise identity provider (SSO) and existing role model at NCA, to align rather than duplicate.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm privileged/step-up authentication requirements per role.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm delegation and emergency-access (break-glass, Ch.36) policies.

26.8 Acceptance Criteria

- A permission check for a past action demonstrably uses the Role/Delegation state effective at that action's timestamp, not the current state.
- Every sampled access/action record across at least three domain chapters demonstrably distinguishes Human, Service, Workflow and ARGUS actor classes.

Audit & Accountability Data Model

Defines the accountable record of who accessed, changed, approved, executed or revealed sensitive NCIE information.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

27.1 Purpose

Defines the accountable record of who accessed, changed, approved, executed or revealed sensitive NCIE information, implementing NCIE-002 Ch.23's Audit Service.

27.2 Semantic Definition

- Read, reveal and export actions can be consequential and are audited exactly like writes where they touch protected/sensitive classification (Ch.31).
- Unknown outcome is a valid, first-class Audit Event state — an action whose result could not be confirmed is recorded as Unknown, never silently omitted or guessed.
- Audit complements but does not replace Provenance (Ch.25) — Audit records who did what and when; Provenance records how a fact was derived.

27.3 Principal Entities

Table 29. Audit & Accountability principal entities.

Entity	Attributes / Attribute Classes
Audit Event	canonical ID, actor reference (Ch.26), target reference, action, timestamp.
Actor	reference to User or Service Identity (Ch.26) — always one of the four distinguishable classes.
Target	the object acted upon (any canonical entity across this document).
Action	governed enumeration (read/reveal/export/create/modify/approve/execute/delete).
Before / After State	audit event reference, prior value, new value (for modify actions).
Reason	audit event reference, stated justification, required for sensitive-access events (Ch.36 break-glass).
Outcome	audit event reference, result (success/failure/unknown).
Correlation ID	audit event reference, cross-system correlation identifier (Ch.28).
Authority Reference	audit event reference, the Role/Delegation/Approval (Ch.26, Ch.23) under which the action was permitted.
Sensitive-Access Marker	audit event reference, flag distinguishing heightened-audit events (protected identity reveal, blocking action, rule change) from routine ones.

27.4 Heightened-Audit Events

Sensitive-Access Marker is set for: Protected Identity Reference reveal (Ch.14), Anti-Fraud sensitive joins (Ch.15), Blocking Action execution (Ch.15), Rule changes (Ch.24), Decision issuance (Ch.16), and material collaborative events (participation-mode change, sensitive evidence introduction, formal promotion, room-originated automation — Ch.21, mirroring NCIE-002 Ch.23 §23.6's Material Collaboration Audit Events).

27.5 Fail-Closed Requirement

If the Audit Service cannot record an Audit Event for an action requiring one, the action itself is blocked from completing rather than proceeding unaudited — this is a data-model consequence of NCIE-002 Ch.23 §23.9's fail-

closed principle, requiring the audit write and the action's own completion to be part of the same transactional boundary where the action is sensitive.

27.6 Relationships and Cardinality

Every Audit Event references exactly one Actor and one Target, and zero-or-one Authority Reference (some routine reads may not require an explicit authority citation beyond ordinary role permission).

27.7 Failure / Exception Semantics

An Audit Event with Outcome Unknown (e.g. the target system's confirmation could not be retrieved) triggers the same Reconciliation State pattern as Chapter 23's workflow outcomes, rather than being left permanently ambiguous.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.23 in full (audit service, heightened-audit events, fail-closed principle).

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm audit retention period against applicable records/regulatory requirements.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which read operations beyond §27.4's list require audit.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm fail-closed requirements for each high-consequence action category.

27.8 Acceptance Criteria

- Every action listed in §27.4 demonstrably produces a Sensitive-Access-Marked Audit Event.
- A simulated Audit Service failure during a sensitive action demonstrably blocks the action rather than letting it proceed unaudited.

Notification, Event & Integration Data Model

Defines internal domain events, event envelopes, subscriptions and notification-delivery state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

28.1 Purpose

Defines internal domain events, event envelopes, subscriptions and notification-delivery state, implementing NCIE-002 Ch.25's Event Bus.

28.2 Semantic Definition

- Sensitive payloads are referenced, not copied, where possible — an event carrying a pointer to a protected Evidence Object (Ch.17) is preferred over one embedding the protected content inline.
- Notification delivery is not business acknowledgement — a Notification's Delivery Status reflects whether the message reached its channel, not whether a human acted on it.

28.3 Principal Entities

Table 30. Notification, Event & Integration principal entities.

Entity	Attributes / Attribute Classes
Event Envelope	canonical ID, event type, schema version (Ch.33), producer reference, correlation ID, published timestamp.
Event Type	governed enumeration (source events, PM/KPI events, Alert/Situation events, Workflow events, collaborative events — per NCIE-002 Ch.25 §25.3).
Producer	service identity reference (Ch.26) that published the event.
Object Reference	event envelope reference, referenced canonical object(s), by pointer rather than embedded copy for sensitive objects.
Correlation ID	shared identifier linking a chain of related events/requests (Ch.23, Ch.27).
Subscription	subscriber reference, event type filter, delivery channel.
Notification	subscription reference, event envelope reference, rendered content.
Delivery Attempt	notification reference, attempt number, timestamp, result.
Delivery Status	notification reference, current status (pending/delivered/failed) — distinct from business acknowledgement (Ch.18's Acknowledgement entity).

28.4 Idempotent Consumption

Every Event Envelope's Correlation ID and an idempotency key allow a consuming service to detect and ignore a duplicate delivery, so a re-delivered event never duplicates its downstream effect (shared mechanism with Ch.23 §23.6).

28.5 Relationships and Cardinality

One Event Envelope may be delivered to zero-to-many Subscriptions, each producing its own Notification with its own Delivery Attempts. A failed Delivery Attempt does not lose the Notification; it is retried per Ch.33's compatibility/retry policy.

28.6 Failure / Exception Semantics

If the Event Bus is degraded, a Producer buffers Event Envelopes locally and retries publication — no event is silently dropped (NCIE-002 Ch.25 §25.7).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.25 in full.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm integration/event platform standards already in use, if any, at NCA.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm external notification channels (email, SMS, dashboard) required at launch.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm which event payloads may leave NCIE's boundary (e.g. to an external notification channel) and under what data-handling policy.

28.7 Acceptance Criteria

- A simulated duplicate event delivery demonstrably does not duplicate downstream effect.
- A sensitive Object Reference is demonstrably passed by pointer, not embedded value, in a sampled Event Envelope.

Data Quality, Validation & Exception Model

Defines how NCIE represents invalid, incomplete, stale, partial or conflicting data.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

29.1 Purpose

Defines how NCIE represents invalid, incomplete, stale, partial or conflicting data, consolidating the Chapter 2 vocabulary into a governed, queryable quality-state model.

29.2 Semantic Definition

- Missing is not zero or normal — this chapter's entities are the enforcement mechanism for Chapter 2's null/unknown vocabulary across every domain.
- Partial coverage remains visible as a first-class state, never rounded up to complete.
- Pipeline/source quality (did the data arrive correctly) is separate from substantive sector condition (is the underlying network/traffic/revenue situation actually good or bad) — a Validation Result never silently stands in for a Network Condition (Ch.7) or Condition (Ch.18).

29.3 Principal Entities

Table 31. Data Quality, Validation & Exception principal entities.

Entity	Attributes / Attribute Classes
Validation Result	See Ch.6 — this chapter extends it with a governed Quality Issue taxonomy.
Quality Issue	validation result reference, issue type (schema/completeness/freshness/conflict), severity.
Completeness State	scope (source/period/domain), expected-vs-received ratio, status (per NCIE-002 Ch.6 §6.9 Green/Amber/Grey convention).
Freshness State	scope, last-updated time, freshness threshold (Ch.24), status (current/stale).
Schema Exception	source object reference (Ch.6), expected schema version (Ch.33), actual structure, deviation.
Source Conflict	two-or-more conflicting Source Object References for the same fact, retained per Ch.5/Ch.17 contradiction-preservation.
Remediation Task	quality issue reference, assignee, status.
Resolution	remediation task reference, outcome, timestamp.

29.4 Quality State Does Not Become Substantive State

A Completeness or Freshness State of Amber/Grey/Stale never itself raises a Network Condition (Ch.7) or REWS Condition (Ch.18) — those remain governed by substantive domain rules (Ch.24). The two kinds of state are cross-referenced for operator context (Ch.37) but never merged.

29.5 Domain-Specific Freshness Thresholds

Freshness State's threshold is a Rule Version (Ch.24), scoped per domain — Network PM data may have a different freshness expectation than Mobile Money aggregates, and each is governed independently rather than sharing one global staleness constant.

29.6 Failure / Exception Semantics

A Schema Exception blocks the affected Source Object Reference from validation (Ch.6) until resolved; it never silently coerces mismatched data into the expected shape.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.6 §6.9 and Ch.28 (source health, data quality) at the platform-wide data-model level.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm domain-specific freshness thresholds per source (Ch.24 Rule Version values).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm data-quality ownership (who is accountable for a given source's Remediation Tasks).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which quality failures block downstream processing entirely versus merely flag it.

29.7 Acceptance Criteria

- A Completeness State of Amber is demonstrably shown alongside, and never in place of, the underlying domain's own substantive status.
- A Schema Exception demonstrably blocks validation rather than silently coercing data.

Metadata Catalogue & Data Dictionary Architecture

Defines the metadata fields that every eventual data-dictionary entry must contain.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

30.1 Purpose

Defines the metadata fields every eventual Data Dictionary entry must contain — the structural standard that Chapter 39's worked entities populate.

30.2 Semantic Definition

The final Data Dictionary must define both business meaning (what a field means to a regulator or analyst) and technical representation (its type, constraints and lineage) for every canonical field. Enumerations require the same governance as Rules (Ch.24) — an enumeration value cannot be added informally.

30.3 Mandatory Dictionary Columns

Table 32. Mandatory Data Dictionary columns (structural standard for Chapter 39).

Column	Definition
Business Name	Human-readable name used in analyst-facing contexts.
Technical Name	Canonical field identifier used in schemas/APIs.
Definition	Authoritative business meaning, unambiguous enough for regulatory defensibility.
Entity	Owning canonical entity (Ch.3 family, specific chapter).
Data Type	Canonical logical data type (not yet a physical column type, Ch.32).
Length / Precision	Where applicable to the logical type.
Format	Encoding/format convention (e.g. ISO 8601 for dates).
Nullability	Whether Unknown/absent is permitted, and what it means if so (Ch.2, Ch.29).
Default	Default value if any — must never silently substitute for Unknown (Ch.2).
Enumeration	Governed value set reference, if applicable.
Owner	Accountable steward for the field's definition.
Source	Canonical NCIE Field vs Source-System Field Mapping distinction (§30.4).
Classification	Tier per Ch.31.
Validation	Constraint(s) enforced, referencing Ch.29's Quality Issue taxonomy where relevant.
Lineage	Reference into Ch.25's Lineage Record for how the field's value is derived.

30.4 Canonical NCIE Field vs Source-System Field Mapping

These are two distinct dictionary layers. A Canonical NCIE Field is defined by NCIE-001/002/003 semantics and may be specified before any particular source system's schema is known. A Source-System Field Mapping records how a specific external system's actual field populates that canonical field, and is only populated once the source system has actually been inspected — it is never fabricated to make the dictionary look complete (§30.5).

30.5 Governance of Enumerations

An enumeration referenced by the Enumeration column is itself a governed reference-data set (versioned like a Rule, Ch.24); adding, renaming or removing a value requires the same change-control discipline as Chapter 1's canonical-definition change process.

30.6 Failure / Exception Semantics

A dictionary entry lacking a confirmed Source-System Field Mapping is marked SOURCE DISCOVERY REQUIRED in that column rather than left blank or guessed (implements the mandatory convention used throughout Chapter 39).

NCIE-002 Architecture Dependency & Traceability

Provides the metadata structure that operationalises NCIE-002's data-model requirements across every domain chapter of this document.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Approve the mandatory dictionary columns in §30.3 as binding for Chapter 39 and any future NCIE-004 implementation-level dictionary.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm naming/abbreviation conventions for Technical Name beyond Chapter 2's baseline rules.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm ownership/stewardship metadata conventions (how an Owner is identified and kept current).

30.7 Acceptance Criteria

- Every worked entity in Chapter 39 demonstrably populates all §30.3 columns, using SOURCE DISCOVERY REQUIRED where genuinely unknown.
- No Chapter 39 entry fabricates a Source-System Field Mapping value.

Classification, Privacy, Retention & Data Lifecycle

Defines how NCIE data is classified, retained, archived, held and securely disposed of.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

31.1 Purpose

Defines how NCIE data is classified, retained, archived, held and securely disposed of — the classification tiers and retention rules referenced throughout this document.

31.2 Semantic Definition

- Protected identity (Ch.14) and precise tracking/location data (Ch.8, Ch.15) require heightened protection beyond ordinary internal classification.
- Retention differs by data class and purpose — a single global retention period is not assumed.
- Derived indexes, embeddings and caches (Ch.20's ARGUS tooling, search) must be considered during lawful deletion — deleting a canonical record without also addressing its derived copies is not lawful deletion.

31.3 Principal Entities

Table 33. Classification, Privacy, Retention & Data Lifecycle principal entities.

Entity	Attributes / Attribute Classes
Classification	tier (public/internal/protected/sensitive), applies-to entity/field reference.
Privacy Category	governed category (e.g. identity document, location, financial), applicable legal basis reference.
Retention Rule	data class, retention period, legal basis, effective period (Ch.24 pattern).
Hold	object reference, reason (e.g. litigation/regulatory hold), effective period, overrides normal retention.
Archive State	object reference, tier (hot/warm/cold, Ch.35), archived timestamp.
Disposition Event	object reference, action (archived/purged), timestamp, authorizing actor (Ch.26).
Deletion Propagation	object reference, derived-copy references (index/embedding/cache) requiring corresponding deletion, status.
Redaction / Masking State	object reference, masked fields, applicable viewer scope.

31.4 Deletion Propagation Requirement

A Disposition Event of type purge for a canonical object is not considered complete until its Deletion Propagation record shows every known derived copy (search index entries, ARGUS-related embeddings, Ch.20; caches) either deleted or explicitly exempted with a recorded reason — this closes the gap NCIE-002 Ch.26 flags around sensitive semantic/vector indexing.

31.5 Hold Overrides Retention

An active Hold on an object suspends its normal Retention Rule outcome; a Disposition Event cannot purge an object with an active Hold, regardless of how expired its retention period otherwise is.

31.6 Classification Inheritance

Consistent with Chapter 2 §2.9, an entity's Classification tier is a floor for its fields and for any derived object referencing it (e.g. an ARGUS Contribution citing sensitive Evidence inherits at least sensitive classification, Ch.20 §20.7).

31.7 Failure / Exception Semantics

A Deletion Propagation that cannot confirm a derived copy was actually removed (e.g. an external cache outside NCIE's direct control) is left in an open, flagged state rather than marked complete on an assumption.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.26 in full (classification, privacy, sovereignty, sensitive-index policy).

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm classification taxonomy against applicable Ghanaian data-protection law and NCA policy.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm retention schedules and legal-hold procedures.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm privacy obligations specific to Ghana Card/Passport-derived data (Ch.14).

31.8 Acceptance Criteria

- A purge Disposition Event for a sampled object with a search-index derived copy demonstrably includes a corresponding Deletion Propagation entry.
- An object under an active Hold is demonstrably blocked from purge regardless of retention-period expiry.

Physical Data Design & Storage Mapping

Maps logical entities into storage technologies after the logical model is approved.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

32.1 Purpose

Maps logical entities into storage technologies after the logical model is approved. This chapter is explicitly downstream of Chapters 1-31; it does not redefine canonical meaning, only where and how it is physically stored.

32.2 Semantic Definition — Three Model Levels

Table 34. The three model levels this document distinguishes throughout; this chapter is the first to address the third.

Model Level	Defines	Chapter(s)
Conceptual Model	Business/regulatory meaning of entities and relationships, technology-agnostic.	Ch.3, domain chapters 7-22
Logical Canonical Model	Entities, attributes, relationships, cardinality, keys — still technology-agnostic.	Ch.4-Ch.30, Ch.40
Physical Implementation Model	Actual database technology, table/collection structures, indexes, partitioning, source-specific mappings.	This chapter, Ch.35

32.3 Principal Entities

Table 35. Physical Data Design & Storage Mapping principal entities.

Entity	Attributes / Attribute Classes
Relational Store	workload class, technology reference, hosted entity families.
Analytical / Time-Series Store	workload class (e.g. PM Observations, Traffic), technology reference.
Object / Document Store	workload class (e.g. Evidence Artifacts), technology reference.
Graph Store	workload class (Ch.19 Intelligence Graph), technology reference.
Search Index	workload class, technology reference, classification-aware segmentation (Ch.31).
Vector Index	workload class (ARGUS retrieval, Ch.20), technology reference, classification-aware segmentation.
Immutable Snapshot / Archive Store	workload class (Ch.25 Snapshots, Ch.35 cold tier), technology reference.
Logical-to-Physical Map	logical entity reference, physical store reference, mapping notes.

32.4 One Technology Is Not Forced Onto Every Workload

The Logical Canonical Model (Ch.4-Ch.30) does not mandate a single database technology; this chapter's Logical-to-Physical Map lets different entity families use the store best suited to their access pattern (e.g. bitemporal relational storage for Revenue, Ch.12; time-series storage for PM Observations, Ch.7; graph storage for the Intelligence Graph, Ch.19) without those choices leaking back into the logical model's meaning.

32.5 Derived vs Canonical Stores

Graph, Search and Vector stores are derived or specialised representations of canonical data (per Ch.31 §31.4's deletion-propagation requirement); none of them is independently canonical, and a discrepancy between a derived store and its canonical source is a data-quality issue (Ch.29), not a competing truth.

32.6 Consistency and Transaction Boundaries

The Logical-to-Physical Map states, for each mapped entity family, its required consistency model (strong/eventual) and transaction boundary (e.g. a Revenue Calculation and its Traffic Input References must be read consistently together) — this must be explicit before physical technology selection, not discovered afterward.

32.7 Failure / Exception Semantics

A derived store found out of sync with its canonical source (Ch.29 Source Conflict pattern applied to derived stores) triggers a Remediation Task (Ch.29) rather than being silently trusted until the next scheduled rebuild.

NCIE-002 Architecture Dependency & Traceability

Depends on NCIE-002 Ch.5 (layered canonical/analytical platform) and Ch.30 (implementation-level persistence, to be detailed in NCIE-014).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm approved enterprise platforms/technologies for each store type in §32.3.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm hosting constraints (on-prem/sovereign cloud, per NCIE-002 Ch.1 §1.6's proposed posture).
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm which stores may contain restricted/sensitive data given each candidate technology's own security posture.

32.8 Acceptance Criteria

- The Logical-to-Physical Map demonstrably covers every entity family in Chapter 3's enterprise model.
- A sampled derived store (Search or Vector Index) is demonstrably reconcilable against its canonical source with a defined remediation path.

Data API, Schema & Event Contract Standards

Defines how data contracts are serialized, versioned and evolved across services.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

33.1 Purpose

Defines how data contracts are serialized, versioned and evolved across services, implementing NCIE-002 Ch.25's Schema Registry at the data-contract level.

33.2 Semantic Definition

A contract requires a stable ID and version independent of its serialization format. Temporal, provenance and classification context (Ch.5, Ch.25, Ch.31) must not be stripped when a canonical fact crosses an integration boundary — a consumer receiving a fact via API or event must still be able to determine its knowledge time, lineage and classification.

33.3 Principal Entities

Table 36. Data API, Schema & Event Contract Standards principal entities.

Entity	Attributes / Attribute Classes
API Schema	schema ID, version, request/response structure, owning service.
Event Schema	schema ID, version, payload structure, owning event type (Ch.28).
Schema ID / Version	stable identifier plus semantic version (major.minor).
Compatibility Policy	schema reference, rule (additive-only for minor, migration-required for major).
Idempotency Key	See Ch.23/Ch.28 — this chapter defines it as a mandatory contract element wherever a call has a side effect.
Correlation ID	See Ch.27/Ch.28 — mandatory contract element for tracing.
Serialization Format	governed reference (e.g. JSON, Protobuf), per schema.
Deprecation State	schema reference, deprecation date, replacement schema reference, sunset date.

33.4 Compatibility Rule

A minor version increment may only add optional fields (additive-first, consistent with NCIE-002 Ch.25 §25.4). Removing or redefining an existing field's meaning requires a major version increment, a Deprecation State on the prior version, and a migration plan (Ch.34) for existing consumers.

33.5 Mandatory Context Fields

Every API Schema and Event Schema that carries a canonical fact includes, at minimum: knowledge time (Ch.5), a lineage reference (Ch.25), and a classification tag (Ch.31) — a schema omitting any of these for a canonical-fact-bearing field is non-conformant and blocked at schema review.

33.6 Relationships

Every Event Type (Ch.28) references exactly one current Event Schema version; historical events retain the schema version they were published under, so a consumer replaying old events can interpret them correctly even after the schema evolves.

33.7 Failure / Exception Semantics

A payload that fails schema validation is treated as a Schema Exception (Ch.29 §29.3), never coerced into the expected shape.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.25 in full (schema registry, compatibility rules).

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm preferred API/event standards already in use at NCA (e.g. existing REST/gRPC/messaging conventions).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm versioning/deprecation policy timelines (how long a deprecated schema remains supported).
- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm external operator contract constraints (e.g. fixed formats operators already use for regulatory submissions).

33.8 Acceptance Criteria

- A sampled schema carrying a canonical fact demonstrably includes knowledge time, lineage reference and classification tag.
- A breaking change is demonstrably rejected unless published as a new major schema version with a Deprecation State on the prior one.

Migration, Legacy Data & Master Data Reconciliation

Defines how existing NCA and operator historical data will be imported and reconciled to canonical NCIE entities.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

34.1 Purpose

Defines how existing NCA and operator historical data will be imported and reconciled to canonical NCIE entities.

34.2 Semantic Definition

- Legacy import is not automatically validated truth — imported records pass through the same Validation Result process (Ch.6, Ch.29) as any other source.
- Original source IDs and transformations are retained through migration, never discarded once a canonical match is found.
- Ambiguous identity matches are never forced during migration, exactly as Chapter 4 §4.4 requires for ongoing identity resolution.

34.3 Principal Entities

Table 37. Migration, Legacy Data & Master Data Reconciliation principal entities.

Entity	Attributes / Attribute Classes
Legacy Source	system name, description, data owner, decommission status.
Legacy Record	legacy source reference, original identifier, original structure (pre-mapping).
Mapping Rule	legacy source reference, field-to-canonical-field mapping (references Ch.30/Ch.39 Source-System Field Mapping).
Transformation	legacy record reference, transformation applied, resulting canonical candidate.
Canonical Match	legacy record reference, candidate canonical entity reference (Ch.4).
Match Confidence / State	canonical match reference, confidence score, state (auto-matched/human-confirmed/ambiguous/rejected).
Migration Batch	scope, source, target, run timestamp, status.
Validation Result	See Ch.6/Ch.29 — migration reuses, not duplicates, this entity.
Exception	migration batch reference, unresolved record reference, reason.

34.4 Migration Flow

Figure 16. Legacy-to-canonical migration and reconciliation flow.

01	Legacy Source inventoried with owner and decommission status
02	Legacy Record extracted, original identifier retained
03	Mapping Rule applies field-to-canonical Transformation
04	Canonical Match attempted via Ch.4's identity-resolution rules
05	Match Confidence/State recorded — ambiguous matches held for human confirmation, never forced
06	Migration Batch completes with Exceptions explicitly listed, not silently dropped

34.5 Provenance Through Migration

A migrated canonical fact retains a Lineage Record (Ch.25) back to its Legacy Record and original Legacy Source identifier permanently — migration is a transformation, not a fresh, unprovenanced canonical fact.

34.6 Failure / Exception Semantics

A Legacy Record that cannot be matched to any canonical entity, and does not meet the bar for creating a new one, becomes an Exception retained for later manual reconciliation, never silently excluded from the migrated dataset.

NCIE-002 Architecture Dependency & Traceability

Depends on NCIE-002 Ch.6 (acquisition/validation reuse) and Ch.5 (canonical entity registry); introduces no new NCIE-002 architecture.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Inventory legacy systems that require migration and their current data owners.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm migration priorities and required historical depth (how far back must legacy data be migrated).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm acceptable manual-reconciliation procedures for Exceptions.

34.7 Acceptance Criteria

- A sampled migrated record demonstrably retains its original Legacy Source identifier and full Transformation lineage.
- An ambiguous legacy match is demonstrably held for human confirmation rather than auto-merged.

Data Performance, Partitioning, Archival & Scalability

Defines how the data layer will handle large current and historical volumes without corrupting semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

35.1 Purpose

Defines how the data layer handles large current and historical volumes without corrupting semantics, implementing NCIE-002 Ch.30's capacity model at the data-design level.

35.2 Semantic Definition

- Hot/Warm/Cold storage tiering changes performance characteristics, never meaning — a Cold-tier canonical fact is exactly as authoritative as the same fact in Hot tier.
- Partitioning must preserve cross-period traceability — a query spanning a partition boundary must still return correct, complete results.
- Optimisation (materialised views, caches) must never mix current and historical versions of the same bitemporal fact (Ch.5).

35.3 Principal Entities

Table 38. Data Performance, Partitioning, Archival & Scalability principal entities.

Entity	Attributes / Attribute Classes
Partition Strategy	entity family reference, partition key (typically operator/geography/time), rationale.
Indexing	entity/field reference, index type, purpose.
Hot / Warm / Cold Tier	entity family reference, tier, transition rule (age/access-frequency based).
Archive Retrieval	archived object reference (Ch.31), retrieval SLA, requesting actor.
Aggregation / Materialization	source entity reference, aggregation logic reference, refresh policy — never itself canonical (Ch.32 §32.5).
Capacity Metadata	entity family reference, current volume, growth rate, projected capacity need.
Performance SLO	entity family/operation reference, target latency/throughput (Ch.30's NCIE-002 targets translated to data-layer terms).

35.4 Tiering Does Not Change Meaning

A Hot/Warm/Cold Tier transition is purely a storage-location and retrieval-latency decision; it never triggers a schema change, a classification change, or a semantic reinterpretation of the underlying canonical fact — those remain governed exclusively by Chapters 2, 5 and 31.

35.5 Partition-Spanning Query Correctness

The Partition Strategy for any entity family bitemporally significant (Ch.5 §5.5) must ensure a decision-time reconstruction (Ch.25) spanning multiple partitions returns a complete, correctly ordered result — partitioning is a physical concern that must not silently truncate a logical query's correctness.

35.6 Failure / Exception Semantics

An Archive Retrieval request that cannot meet its stated SLA is surfaced as a degraded-status response, never a silent delay presented as normal performance.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.30 in full (capacity model, performance targets) at the data layer.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Confirm expected data volumes and retention horizons per entity family.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm performance targets (translate NCIE-002 Ch.30 §30.2's proposed baseline into per-entity-family SLOs).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm archive-retrieval SLA expectations for Cold-tier data.

35.7 Acceptance Criteria

- A decision-time reconstruction spanning a partition boundary is demonstrably complete and correctly ordered.
- A Hot-to-Cold tier transition is demonstrably invisible to a consumer querying by canonical meaning alone.

Data Security & Sovereignty Controls

Defines technical data-layer protections for confidentiality, integrity and jurisdictional control.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

36.1 Purpose

Defines technical data-layer protections for confidentiality, integrity and jurisdictional control, implementing NCIE-002 Ch.26 at the data-model level.

36.2 Semantic Definition

- Encryption and masking are distinct controls — encryption protects data at rest/in transit; masking/tokenization controls what a given viewer sees regardless of encryption state.
- Sensitive semantic/vector indexing (Ch.20's ARGUS retrieval, Ch.32's Vector Index) requires design distinct from ordinary indexing, since embedding sensitive content can make it searchable in ways ordinary access control does not anticipate.
- Data residency follows NCA policy — no entity in this document assumes cross-border processing as a default.

36.3 Principal Entities

Table 39. Data Security & Sovereignty Controls principal entities.

Entity	Attributes / Attribute Classes
Encryption	object/store reference, algorithm, key reference.
Key Reference	key identifier — actual key material is never stored alongside the data it protects (managed per NCIE-002 Ch.26 §26.7).
Masking / Tokenization	field reference, masking method, applicable viewer scope (Ch.26 of this document).
Protected Reveal	masked field reference, revealing actor, authorization reference (Ch.26), audited (Ch.27).
Database Access Policy	store reference, permitted access pattern, enforcement point.
Sovereignty / Residency Metadata	store reference, jurisdiction, hosting constraint.
DLP Control	scope, rule (e.g. block export of protected identity in bulk), enforcement point.
Sensitive-Index Policy	index reference (Ch.32), classification-aware segmentation rule, exclusion list.

36.4 Sensitive-Index Design Requirement

A Vector Index or Search Index that could include protected/sensitive content must apply Sensitive-Index Policy segmentation (index-per-classification-tier, security metadata filtering, or post-retrieval authorization, per NCIE-002 Ch.26 §26.3) — embedding a protected field is never permitted to make that content globally searchable to any authenticated user.

36.5 Protected Reveal as an Audited Action

Every instance of a masked field being shown unmasked to a viewer is a Protected Reveal record, which is itself an Audit Event (Ch.27) with a Sensitive-Access Marker — masking is not considered a security control unless every bypass of it is individually accountable.

36.6 Failure / Exception Semantics

If Key Reference resolution fails (key management unavailable), an operation requiring decryption of protected/sensitive data fails closed rather than falling back to an unencrypted path (NCIE-002 Ch.28's parallel resilience principle applied here).

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.26 in full.

Human Review Focus

- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm sovereignty/residency requirements with NCA legal/IT security leadership.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm encryption/key-management standards and rotation policy.
- **[LEGAL/POLICY CONFIRMATION REQUIRED]** Confirm which data classes may use external/cloud processing, if any.

36.7 Acceptance Criteria

- A protected field embedded in a Vector Index is demonstrably not retrievable by a role unauthorized for that field.
- Every sampled Protected Reveal demonstrably produces a corresponding Sensitive-Access-Marked Audit Event.

Data Observability, Reconciliation & Operational Management

Defines operational telemetry for freshness, pipeline health, schema drift and reconciliation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

37.1 Purpose

Defines operational telemetry for freshness, pipeline health, schema drift and reconciliation, implementing NCIE-002 Ch.27's Observability Platform at the data layer.

37.2 Semantic Definition

- Pipeline health (is ingestion working) is not sector health (is the underlying network/revenue/fraud situation good) — this chapter's entities feed, but never substitute for, Ch.7/Ch.18's substantive condition entities.
- Source update time and processing time are kept distinct so a lag can be attributed to the source or to NCIE's own pipeline correctly.

37.3 Principal Entities

Table 40. Data Observability, Reconciliation & Operational Management principal entities.

Entity	Attributes / Attribute Classes
Pipeline Status	pipeline reference, current state, last-run outcome.
Freshness Metric	See Ch.29 — this chapter's operational view over Freshness State.
Ingestion Lag	source reference (Ch.6), source update time, NCIE processing time, computed lag.
Schema Drift Event	source reference, expected schema version (Ch.33), detected deviation, timestamp.
Lineage Exception	See Ch.25 — broken/incomplete lineage surfaced as an operational alert.
Reconciliation Job	scope, schedule, last-run result — the operational counterpart to Ch.23/Ch.27's Reconciliation State.
Operational Alert	category, severity, triggering condition, escalation reference (Ch.18 §18.4 pattern reused operationally).
Data SLA / SLO	entity family/pipeline reference, target, current attainment.

37.4 Operational vs Substantive Signals

An Operational Alert for a data-pipeline issue is routed to data engineering/operations owners (Ch.26), while a substantive Condition/Alert (Ch.18) is routed to domain analysts — the two alerting paths share infrastructure (Ch.28's Event/Notification model) but are never merged into one undifferentiated alert stream.

37.5 Reconciliation Scheduling

A Reconciliation Job runs on a defined schedule per entity family (e.g. daily for Revenue inputs, per Ch.12) and produces a result that feeds both Ch.29's Quality Issue tracking and this chapter's Data SLA/SLO attainment metric.

37.6 Failure / Exception Semantics

A Schema Drift Event blocks affected Source Object References from validation (shared mechanism with Ch.29 §29.6) and raises an Operational Alert simultaneously, so the pipeline issue is visible to operators even before an analyst notices missing domain data.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.27 in full (observability platform, health signal categories).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm operational owners and escalation paths per pipeline.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm freshness expectations per domain (feeds Ch.24 Rule Version thresholds).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm acceptable reconciliation windows per entity family.

37.7 Acceptance Criteria

- A simulated Schema Drift Event demonstrably raises an Operational Alert distinct from any substantive domain Condition/Alert.
- A Reconciliation Job's result is demonstrably reflected in both Ch.29's Quality Issue tracking and this chapter's Data SLA/SLO metric.

Data Testing, Verification & Acceptance

Defines how the data architecture and dictionary will be verified before implementation baseline approval.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

38.1 Purpose

Defines how the data architecture and dictionary are verified before implementation baseline approval, implementing NCIE-002 Ch.31's test-architecture principles at the data-model level.

38.2 Semantic Definition

Testing here covers semantics, not only database constraints — a passing schema-validation test does not by itself demonstrate that temporal, lineage, or authorization behaviour is correct. Temporal/Decision-Time behaviour, security/lineage and migration each require dedicated test categories, per the authoritative handover instructions.

38.3 Principal Entities

Table 41. Data Testing, Verification & Acceptance principal entities.

Entity	Attributes / Attribute Classes
Schema Tests	verify structural conformance to Ch.33 contracts.
Constraint Tests	verify nullability, enumeration, classification-tier rules (Ch.2, Ch.30, Ch.31).
Temporal Tests	verify bitemporal storage, correction/revision behaviour and effective dating (Ch.5, Ch.24).
Lineage Tests	verify Lineage Record completeness and decision-time reconstruction correctness (Ch.25).
Authorization / Privacy Tests	verify field-level, room-level and classification-aware access rules (Ch.26, Ch.21, Ch.31, Ch.36).
Migration Tests	verify legacy-to-canonical mapping, provenance retention and ambiguous-match handling (Ch.34).
Reconciliation Tests	verify Ch.37's reconciliation jobs and Ch.29's quality-issue detection.
Acceptance Evidence	test case reference, executed-by, date, result, defect reference — shared structure with NCIE-002 Ch.31.

38.4 Mandatory Test Categories

Every domain chapter's principal entities (Ch.7-Ch.22) must have at least one Temporal Test (bitemporal correctness), one Authorization/Privacy Test (field/classification enforcement) and one Lineage Test (provenance completeness) before that chapter is eligible for Approved status (front matter §0.3).

38.5 Relationship to NCIE-002 Acceptance

This chapter's Acceptance Evidence entity shares its structure with NCIE-002 Ch.31's RTM acceptance record, so a single evidence repository can serve both documents' traceability needs without duplication.

38.6 Failure / Exception Semantics

A failed mandatory test category for a chapter blocks that chapter's status from advancing beyond IN DEVELOPMENT — FOR HUMAN REVIEW, mirroring NCIE-001 §18.242's no-pass-without-corrective-evidence rule.

NCIE-002 Architecture Dependency & Traceability

Directly implements NCIE-002 Ch.31 in full, adapted to data-model-specific test categories.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm acceptance authorities for NCIE-003 test results.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm required test environments and data (masked, per NCIE-002 Ch.29 §29.3 convention).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which domain users must sign off on which chapters' Acceptance Evidence.

38.7 Acceptance Criteria

- Every domain chapter (Ch.7-Ch.22) demonstrably has at least one Temporal, one Authorization/Privacy and one Lineage Test recorded.
- No chapter is shown as Approved without corresponding passing Acceptance Evidence.

Canonical Data Dictionary

Holds the authoritative Data Dictionary structure and representative fully worked canonical entities.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

39.1 Purpose

Holds the authoritative Data Dictionary structure (Chapter 30) and representative, fully worked canonical entities demonstrating exactly how the complete dictionary must eventually be populated. Per the authoritative handover instructions, this chapter does not fabricate the production field-level dictionary before NCA data discovery — it builds the complete architecture and structure now, and worked examples across every mandated domain, while marking unknown source-system mappings SOURCE DISCOVERY REQUIRED rather than inventing them.

39.2 How to Read a Worked Entity

Each worked entity below shows: an entity card (business name, technical name, definition, owning chapter, classification, owner) and an attribute table (Attribute, Definition, Data Type, Nullable, Classification, Source Mapping). Source Mapping is populated only where NCIE-001/002/003 already fixes it (e.g. a governed enumeration or an internally computed field); every attribute whose value would depend on inspecting an actual NCA/operator system is marked SOURCE DISCOVERY REQUIRED. Canonical NCIE Field and Source-System Field Mapping (Ch.30 §30.4) are always kept visually distinct.

39.3 Shared/Core — Operator

Table 42. Worked entity: Operator (Shared/Core).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier (Ch.4).	Identifier	No	Internal	Internally assigned
name	Operator's registered trading name.	String	No	Internal	SOURCE DISCOVERY REQUIRED
licence_reference	Reference to the operator's NCA licence record.	String	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED
status	Active / suspended / revoked.	Enumeration	No	Internal	Governed enumeration (Ch.24)

39.4 Network — PM Observation

Table 43. Worked entity: PM Observation (Network).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
cell_reference	Canonical Cell (Ch.7) this observation belongs to.	Reference	No	Internal	Resolved via Topology Service (Ch.8)
kpi_reference	Governed KPI definition (Ch.24).	Reference	No	Internal	Governed reference data
period	Observation interval.	Temporal range	No	Internal	SOURCE DISCOVERY REQUIRED
value	Measured KPI value.	Decimal	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
coverage_status	Green/Amber/Grey per Ch.6 §6.9.	Enumeration	No	Internal	Computed by NCIE, not source-supplied

39.5 Traffic — Traffic Observation

Table 44. Worked entity: Traffic Observation (Traffic).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
operator_reference	Canonical Operator.	Reference	No	Internal	Resolved via Ch.4
traffic_type	International / Off-Net / On-Net.	Enumeration	No	Internal	Governed enumeration (Ch.24)
direction	Incoming/outgoing where applicable.	Enumeration	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED
period	Reporting interval.	Temporal range	No	Internal	SOURCE DISCOVERY REQUIRED
volume	Traffic volume.	Decimal	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED
unit	Unit of measure.	Enumeration	No	Internal	SOURCE DISCOVERY REQUIRED

39.6 Revenue — Revenue Calculation

Table 45. Worked entity: Revenue Calculation (Revenue) — note the surcharge rate itself is never a literal field here; it is looked up via surcharge_rule_version (Ch.24 §24.2).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
operator_reference	Canonical Operator.	Reference	No	Internal	Resolved via Ch.4
period	Calculation period.	Temporal range	No	Internal	Derived from Traffic Input References
traffic_input_references	Exact Traffic Observation version(s) used.	Reference list	No	Internal	Internally computed, immutable pointer (Ch.5 §5.6)
surcharge_rule_version	Effective Rule Version applied (Ch.24).	Reference	No	Internal	Internally resolved from Rule Service
calculated_value	Deterministically calculated Revenue.	Decimal (currency)	No	Internal	Internally computed — never source-supplied
currency	Currency of calculated_value.	Enumeration	No	Internal	SOURCE DISCOVERY REQUIRED

39.7 Mobile Money — Mobile Money Aggregate

Table 46. Worked entity: Mobile Money Aggregate — no subscriber-level field exists in this schema by design (Ch.13 §13.4).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
operator_service_reference	Canonical Operator/Service.	Reference	No	Internal	Resolved via Ch.4
reporting_date	Aggregate date.	Date	No	Internal	SOURCE DISCOVERY REQUIRED
sent_amount	Aggregate sent value.	Decimal (currency)	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED
received_amount	Aggregate received value.	Decimal (currency)	Yes (Unknown, Ch.2)	Internal	SOURCE DISCOVERY REQUIRED
currency	Currency of sent/received amounts.	Enumeration	No	Internal	SOURCE DISCOVERY REQUIRED

39.8 SIM Registration — Registration

Table 47. Worked entity: Registration (SIM Registration).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
sim_reference	Canonical SIM.	Reference	No	Internal	SOURCE DISCOVERY REQUIRED
protected_identity_reference	Protected identity (Ch.4/Ch.14) — value itself lives in the field-level-protected entity, never inline here.	Reference	No	Protected	SOURCE DISCOVERY REQUIRED
identity_type	Ghana Card / Passport.	Enumeration	No	Internal	Governed enumeration
status	Registration status.	Enumeration	No	Internal	SOURCE DISCOVERY REQUIRED
effective_date	Registration effective date.	Date	No	Internal	SOURCE DISCOVERY REQUIRED

39.9 Anti-Fraud/SIMBOX — SIMBOX Detection

Table 48. Worked entity: SIMBOX Detection (Anti-Fraud/SIMBOX) — a Detection alone never asserts Confirmed Fraud (Ch.15 §15.2).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
detection_signal_type	Type of automated signal.	Enumeration	No	Sensitive	SOURCE DISCOVERY REQUIRED
confidence	Detection confidence score.	Decimal	Yes (Unknown, Ch.2)	Sensitive	SOURCE DISCOVERY REQUIRED
detected_time	When the signal was generated.	Timestamp	No	Sensitive	SOURCE DISCOVERY REQUIRED
source_reference	Originating Source (Ch.6).	Reference	No	Sensitive	Resolved via Ch.6

39.10 Evidence — Evidence Object

Table 49. Worked entity: Evidence Object (Evidence).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Varies (inherits content classification)	Internally assigned
source_reference	Originating Source Object Reference (Ch.6).	Reference	Yes (deliberately registered, Ch.17 §17.2)	Varies	Resolved via Ch.6
status	Active / invalidated.	Enumeration	No	Varies	Internally managed
hash_value	Integrity hash of the Evidence Artifact.	String (hash)	No	Varies	Internally computed
classification_tier	Ch.31 classification tier.	Enumeration	No	Varies	Assigned per Ch.31 rules

39.11 REWS/Situation — Alert

Table 50. Worked entity: Alert (REWS/Situation).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
condition_reference	Triggering Condition (Ch.18).	Reference	No	Internal	Internally computed
alert_rule_reference	Effective Alert Rule version (Ch.24).	Reference	No	Internal	Internally resolved
severity	Alert severity.	Enumeration	No	Internal	Governed enumeration, effective-dated
raised_time	When the Alert was raised.	Timestamp	No	Internal	Internally computed

39.12 ARGUS — ARGUS Contribution

Table 51. Worked entity: ARGUS Contribution (ARGUS) — no field for private model chain-of-thought exists anywhere in this entity (Ch.20 §20.2).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Varies (inherits referenced Evidence classification)	Internally assigned
interaction_reference	Parent ARGUS Interaction.	Reference	No	Varies	Internally recorded
contribution_type	Question/Challenge/Hypothesis/Contradiction/Evidence Gap.	Enumeration	No	Varies	Governed enumeration (Ch.20 §20.3)
ai_origin_marker	Always true for this entity — retained even after human modification (Ch.20 §20.4).	Boolean	No	Varies	Internally set, immutable
evidence_references	Cited Evidence Objects (Ch.17) — citation only, not promotion to Evidence.	Reference list	Yes	Varies	Internally recorded
model_route_version	Model/provider and version used (Ch.24).	Reference	No (where governance requires audit)	Internal	Internally recorded from Model Gateway

39.13 Collaborative Intelligence — Hypothesis

Table 52. Worked entity: Hypothesis (Collaborative Intelligence) — never itself a Finding (Ch.21 §21.2).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Varies (Room classification, Ch.31)	Internally assigned
thread_reference	Parent Thread (Ch.21) — prevents cross-thread contamination.	Reference	No	Varies	Internally recorded
originator	Human user or ARGUS (Ch.20) — permanently retained.	Reference	No	Varies	Internally recorded, immutable
content	Hypothesis text.	Text	No	Varies	Internally recorded
supporting_evidence_references	Ch.17 Evidence Objects supporting this hypothesis.	Reference list	Yes	Varies	Internally recorded
status	Active/adopted/deferred/rejected.	Enumeration	No	Varies	Internally managed

39.14 Memory/Institutional Knowledge — Institutional Knowledge

Table 53. Worked entity: Institutional Knowledge (Memory/Institutional Knowledge) — never itself a Rule (Ch.22 §22.2).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Varies (Applicability Scope-derived)	Internally assigned
knowledge_candidate_reference	Source Knowledge Candidate that was promoted.	Reference	No	Varies	Internally recorded
content	Institutional Knowledge content.	Text	No	Varies	Internally recorded
applicability_scope	Domain/geography/time window this applies to.	Structured	No	Varies	Internally recorded
review_state	Reviewer, decision, timestamp — proves human promotion (Ch.22 §22.4).	Structured	No	Varies	Internally recorded

39.15 Workflow — Workflow Instance

Table 54. Worked entity: Workflow Instance (Workflow).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
workflow_definition_reference	Which workflow template this instantiates.	Reference	No	Internal	Internally recorded
originator	Human, Room (Ch.21) or schedule that requested it.	Reference	No	Internal	Internally recorded
state	Requested/Approved/Executed/Verified/Failed/Reconciling.	Enumeration	No	Internal	Governed enumeration (Ch.23 §23.2)

39.16 Rules — Rule Version

Table 55. Worked entity: Rule Version (Rules) — this is where the surcharge rate/SIM limits actually live; no other entity in this dictionary stores them.

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal	Internally assigned
rule_reference	Parent Rule (rule key, e.g. international-surcharge).	Reference	No	Internal	Internally defined
parameters	Named Parameter values (e.g. rate, currency, or limit and identity type).	Structured	No	Internal	Institutionally confirmed values (Ch.24 §24.9 review)
effective_from	Effective start date.	Date	No	Internal	Institutionally confirmed
effective_to	Effective end date (nullable for open-ended).	Date	Yes	Internal	Institutionally confirmed
approval_state	Approving actor and timestamp.	Structured	No	Internal	Internally recorded (Ch.26 actor)

39.17 Audit — Audit Event

Table 56. Worked entity: Audit Event (Audit).

Attribute	Definition	Data Type	Nullable	Classification	Source Mapping
canonical_id	Stable NCIE canonical identifier.	Identifier	No	Internal (append-only)	Internally assigned
actor_reference	User or Service Identity (Ch.26) — one of four distinguishable classes.	Reference	No	Internal	Internally recorded
target_reference	Object acted upon.	Reference	No	Internal	Internally recorded
action	Read/reveal/export/create/modify/approve/execute/delete.	Enumeration	No	Internal	Governed enumeration
sensitive_access_marker	Whether this is a heightened-audit event (Ch.27 §27.4).	Boolean	No	Internal	Internally computed from action/target classification

NCIE-002 Architecture Dependency & Traceability

Consolidates worked examples for entities defined in Ch.3-Ch.30 of this document, which in turn trace individually to their respective NCIE-002 chapters.

Human Review Focus

- **[SOURCE/DATA DISCOVERY REQUIRED]** Every attribute marked SOURCE DISCOVERY REQUIRED above requires actual NCA/operator system inspection before NCIE-004 implementation begins.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm which additional entities beyond these 15 representative examples require full worked treatment before NCIE-003 is considered dictionary-complete.

39.18 Acceptance Criteria

- Every one of the 15 mandated domains has at least one fully worked entity per §39.3-§39.17.

- No worked entity above contains a fabricated Source-System Field Mapping value; every unknown mapping reads SOURCE DISCOVERY REQUIRED.

Entity Relationship Catalogue & Reference Diagrams

Provides authoritative conceptual and logical ER diagrams and relationship catalogues that agree with the Data Dictionary.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

40.1 Purpose

Provides authoritative conceptual and logical ER diagrams and relationship catalogues that agree with Chapter 39's Data Dictionary. Per the hybrid principle, logical canonical ER models are produced now, since NCIE-001/002 already establish enough conceptual architecture; physical table structures, database-specific keys, indexes and partitioning are explicitly out of scope here and belong to Chapter 32 and NCIE-014.

40.2 Model-Level Convention

Every diagram below is explicitly a Logical Canonical Model (Ch.32 §32.2): it shows entities and relationships with cardinality intent, independent of storage technology. None of these diagrams is a Physical Implementation Model — no diagram implies a specific table, key or index structure.

40.3 Enterprise Core ERD

Figure 17. Enterprise Core logical ERD — shared/core entities referenced across domains.

SOURCE	RELATIONSHIP	TARGET
Operator	is referenced by	Cell, Traffic Observation, Revenue Calculation, Registration
Geography	is referenced by	Cell, Incident, Campaign, Condition
Canonical ID (Ch.4)	underlies identity of	every entity in this catalogue
Evidence Object	is referenced by	Finding, Hypothesis, ARGUS Contribution, Graph Edge

40.4 Network ERD

Figure 18. Network logical ERD.

SOURCE	RELATIONSHIP	TARGET
Operator	has	Site
Site	has	Cell
Cell	has many	PM Observation
PM Observation	is compared to	Baseline / Threshold
Degradation pattern	raises	Network Condition

40.5 Traffic ERD

Figure 19. Traffic logical ERD.

SOURCE	RELATIONSHIP	TARGET
Operator	has many	Traffic Observation
Traffic Observation	may have	Revision (Correction relationship, Ch.25)
Traffic Observation	is input to	Revenue Calculation

40.6 Revenue ERD

Figure 20. Revenue logical ERD.

SOURCE	RELATIONSHIP	TARGET
Traffic Observation	feeds	Revenue Calculation

SOURCE	RELATIONSHIP	TARGET
Rule Version (surcharge)	governs	Revenue Calculation
Revenue Calculation	is compared in	Reconciliation
Billing Verification	is compared in	Reconciliation
Reconciliation	may produce	Reconciliation Difference

40.7 Mobile Money ERD

Figure 21. Mobile Money logical ERD (deliberately minimal, no subscriber-level entity).

SOURCE	RELATIONSHIP	TARGET
Operator/Service	has many	Mobile Money Aggregate
Mobile Money Aggregate	is correlated by	Cross-Domain Fusion (Ch.19)

40.8 SIM Registration ERD

Figure 22. SIM Registration logical ERD.

SOURCE	RELATIONSHIP	TARGET
Protected Identity Reference	has many	Registration
Registration	is evaluated for Countable State against	Registration Rule (Ch.24)
Count over limit	raises	Registration Exception
Registration Exception	may become	Case (Ch. 16)

40.9 Anti-Fraud / SIMBOX ERD

Figure 23. Anti-Fraud / SIMBOX logical ERD.

SOURCE	RELATIONSHIP	TARGET
SIMBOX Detection	identifies	Suspected SIM
Suspected SIM	is examined in	Investigation
Investigation	may gather	Location Evidence (under Tracking Authorization)
Investigation	may raise	Blocking Request
Blocking Request	leads to	Blocking Action / Blocking Verification

40.10 Evidence / Investigation ERD

Figure 24. Evidence / Investigation logical ERD.

SOURCE	RELATIONSHIP	TARGET
Source Object Reference	may be registered as	Evidence Object
Evidence Object	has	Hash/Integrity Metadata
Evidence Object	supports or contradicts	Hypothesis / Finding
Evidence Object	may have	Correction (new version) or Invalidation

40.11 REWS / Situation / NOP ERD

Figure 25. REWS / Situation / National Operating Picture logical ERD.

SOURCE	RELATIONSHIP	TARGET
Observation/Condition	raises	Alert
Alert	has	Acknowledgement / Escalation
related Alerts/Incidents/Evidence	are grouped into	Situation
current status across domains	composes	NOP Projection

40.12 ARGUS / Collaborative Intelligence ERD

Figure 26. ARGUS / Collaborative Intelligence logical ERD.

SOURCE	RELATIONSHIP	TARGET
ARGUS Interaction	produces	ARGUS Contribution
ARGUS Contribution	cites	Evidence Object (never becomes one)
Collaborative Room	has	Thread
Thread	has	Hypothesis, Contradiction, Evidence Gap, Task
Room	captures	Room Snapshot

40.13 Memory / Institutional Knowledge ERD

Figure 27. Memory / Institutional Knowledge logical ERD.

SOURCE	RELATIONSHIP	TARGET
Memory Object	may generalise to	Knowledge Candidate
Knowledge Candidate	requires human Review State to become	Institutional Knowledge
Institutional Knowledge	may be	Superseded / Retracted (history retained)

40.14 Workflow / Rules ERD

Figure 28. Workflow / Rules logical ERD.

SOURCE	RELATIONSHIP	TARGET
Workflow Instance	requires	Approval (distinct actor)
Workflow Instance	executes via	Automation/RPA Run
Automation/RPA Run	produces	Outcome (with Reconciliation if Unknown)
Rule	has many	Rule Version (effective-dated)

40.15 Audit / Provenance ERD

Figure 29. Audit / Provenance logical ERD.

SOURCE	RELATIONSHIP	TARGET
Actor (Ch.26)	performs	Audit Event
Audit Event	targets	any canonical object
Lineage Record	links	derived object to originating Source Object Reference
Correction Relationship	flags	Dependency Edge for Impact-Review

40.16 Diagram-Dictionary Agreement

No diagram in this chapter introduces an entity absent from Chapter 3's family catalogue or Chapters 7-30's domain chapters; conversely, every entity family in Chapter 3 appears in at least one diagram here. This mutual coverage is a checked acceptance criterion (§40.18), not an assumption.

NCIE-002 Architecture Dependency & Traceability

Consolidates the entity/relationship structures from Ch.3-Ch.30, agreeing with NCIE-002's domain and platform architecture chapters throughout.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm preferred diagram notation for the eventual formal ERD deliverable (this chapter's relationship-catalogue format is a content-complete stand-in, not a specific notation standard such as Crow's Foot or IE).
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the level of detail appropriate for architectural review versus what NCIE-004/implementation will need.

- **[SOURCE/DATA DISCOVERY REQUIRED]** Validate each domain's relationships with the relevant domain users before physical implementation begins.

40.17 Physical Model Deferral

Table/collection names, primary/foreign keys, indexes and partitioning are Physical Implementation Model concerns (Ch.32, Ch.35) and are deliberately not finalized in this chapter — doing so before technology selection and source discovery would let physical convenience silently redefine canonical meaning, which Chapter 32 §32.4 prohibits.

40.18 Acceptance Criteria

- Every entity family in Chapter 3 §3.3 appears in at least one ERD in this chapter.
- No ERD in this chapter references an entity absent from the Chapter 39 dictionary or Chapters 7-30.

Requirements Traceability & NCIE-002 Crosswalk

Maps every significant data object and field back to the requirements and architecture that justify it.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

41.1 Purpose

Maps every significant NCIE-003 chapter back to the NCIE-002 architecture chapter(s) that justify it, consolidating the per-chapter 'NCIE-002 Architecture Dependency & Traceability' sections that appear throughout Chapters 1-40 (per the authoritative instruction that traceability is mandatory throughout, not postponed to this chapter — this chapter consolidates rather than originates it).

41.2 Master Crosswalk

Table 57. Master crosswalk: every NCIE-003 chapter to its primary NCIE-002 architecture dependency.

NCIE-003 Chapter	Primary NCIE-002 Dependency
1 — Governance & Mandate	Ch.1 §1.3.3 (evidence/provenance-first), Ch.5
2 — Principles & Standards	Ch.1 §1.3, Ch.5, Ch.18 §18.5, Ch.26
3 — Enterprise Canonical Model	Ch.5, Ch.17
4 — Common Entity/Identifier Model	Ch.5 §5.2, Ch.17, Ch.19 §19.11
5 — Temporal & Bitemporal Model	Ch.5 §5.3, Ch.22 §22.3, Ch.23 §23.5
6 — Source/Acquisition Model	Ch.6 (in full)
7 — Network Intelligence Model	Ch.7 (in full)
8 — Topology/GIS/Spatial Model	Ch.8 (in full)
9 — QoS Campaign Model	Ch.9 (in full)
10 — Incident Intelligence Model	Ch.10 (in full)
11 — Traffic Intelligence Model	Ch.11 (in full)
12 — Revenue/Billing Model	Ch.12 (in full); depends on Ch.24 (this document)
13 — Mobile Money Model	Ch.13 (in full)
14 — SIM Registration Model	Ch.14 (in full); depends on Ch.4, Ch.24 (this document)
15 — Anti-Fraud/SIMBOX Model	Ch.15 (in full)
16 — Regulatory Case/Finding/Decision Model	Ch.16 (in full); depends on Ch.20, Ch.26 (this document)
17 — Evidence Model	Ch.23 (in full)
18 — REWS/Alert/Situation/NOP Model	Ch.18 (in full); depends on Ch.24 (this document)
19 — Cross-Domain Fusion/Graph Model	Ch.17 (in full); depends on Ch.20 (this document)
20 — ARGUS Interaction/AI Provenance Model	Ch.19 (in full, incl. §19.11), Ch.24 (in full)
21 — Collaborative Intelligence Model	Ch.20 (in full)
22 — Memory/Institutional Knowledge Model	Ch.20 §20.2, §20.12
23 — Workflow/Automation/RPA Model	Ch.21 (in full)
24 — Rules/Configuration/Calculation Model	Ch.22 (in full)
25 — Provenance/Lineage/Decision-Time Model	Ch.23 (in full)
26 — Identity/Authorization/Security Model	Ch.4 (in full)
27 — Audit/Accountability Model	Ch.23 (in full)
28 — Notification/Event/Integration Model	Ch.25 (in full)
29 — Data Quality/Validation Model	Ch.6 §6.9, Ch.28

NCIE-003 Chapter	Primary NCIE-002 Dependency
30 — Metadata Catalogue Architecture	Operationalises Ch.2/Ch.5 across every domain chapter
31 — Classification/Privacy/Retention	Ch.26 (in full)
32 — Physical Data Design	Ch.5 (canonical/analytical platform), Ch.30 (implementation persistence, → NCIE-014)
33 — Data API/Schema/Event Contracts	Ch.25 (in full)
34 — Migration/Legacy Reconciliation	Ch.6, Ch.5 — no new architecture introduced
35 — Performance/Partitioning/Archival	Ch.30 (in full)
36 — Data Security/Sovereignty	Ch.26 (in full)
37 — Data Observability/Reconciliation	Ch.27 (in full)
38 — Data Testing/Verification/Acceptance	Ch.31 (in full)
39 — Canonical Data Dictionary	Consolidates Ch.3-Ch.30 of this document
40 — ER Catalogue	Consolidates Ch.3-Ch.30 of this document

41.3 Orphan Detection

An orphan field is one present in Chapter 39's dictionary with no traceable justification in either NCIE-001 or NCIE-002; an unrealised requirement is an NCIE-001/002 requirement with no corresponding NCIE-003 entity/field. Neither condition exists in the current chapter set at this baseline — every worked entity in Chapter 39 and every entity family in Chapter 3 originates from a cited NCIE-002 chapter in §41.2.

41.4 Stable Traceability IDs

This document's own stable requirement-ID scheme mirrors NCIE-001's: a chapter.section reference (e.g. NCIE-003 §14.4) is the stable ID used by Chapter 38's test cases and any future NCIE-004/implementation backlog item that needs to cite a specific NCIE-003 decision.

41.5 Upstream Impact Log

No UPSTREAM ARCHITECTURE IMPACT — REVIEW REQUIRED flag has been raised during this expansion; NCIE-003 has been produced entirely by implementing NCIE-002 semantically, without requiring any change to NCIE-002's architecture. This log remains open for any future expansion pass that does identify such an impact.

NCIE-002 Architecture Dependency & Traceability

This chapter's entire content IS the consolidated NCIE-002 crosswalk; see §41.2.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm traceability granularity required (chapter-level, as here, versus field-level for every dictionary entry).
- **[RESOLVED BY APPROVED NCIE BASELINE]** Authoritative NCIE-002 baseline for this crosswalk: NCIE-002 System Architecture & Technical Design, Version 0.4 (Production Ready) — resolved by this edition's front matter §0.1.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Approve the process for resolving any future orphan requirement or field, should one be discovered as the dictionary is populated with real source mappings.

41.6 Acceptance Criteria

- Every row in §41.2 is independently verifiable against the cited NCIE-002 chapter.
- No orphan field or unrealised requirement is found on inspection of Chapters 3-40 against NCIE-001/002.

Open Institutional Decisions & Human Review Register

Captures data decisions that architecture cannot legitimately settle without NCA confirmation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

42.1 Purpose

Consolidates every still-open Human Review Focus item from Chapters 1-40 into one authoritative register, so NCA reviewers can review unresolved decisions in one place rather than searching the full document. Items already labelled RESOLVED or PROPOSED remain in their originating chapters and are not duplicated here in full; this register focuses on INSTITUTIONAL, SOURCE/DATA DISCOVERY and LEGAL/POLICY items, which are the categories that cannot be closed by architecture alone.

42.2 Register Structure

Table 58. Open Institutional Decisions & Human Review Register structure.

Column	Definition
Decision ID	Stable reference (chapter.sequence, e.g. 14-A).
Issue	The open question.
Affected Chapter/Entity	Where in NCIE-003 this decision has effect.
Category	INSTITUTIONAL / SOURCE_DISCOVERY / LEGAL.
Options	Plausible resolutions, where known.
Owner	Proposed accountable party (NCA function), pending formal assignment.
Required Evidence	What would close the item (e.g. an NCA policy document, a source-system inspection).
Status	Open / assigned / closed.
Decision	Recorded once closed.
Effective Date	Recorded once closed.

42.3 Consolidated High-Priority Items

Table 59. High-priority consolidated open decisions (representative; the complete set is the union of every chapter's Human Review Focus section).

Decision ID	Issue	Affected Chapter	Category
1-A	Approval authority for NCIE-003 as a whole and per chapter.	Ch.1	INSTITUTIONAL
2-A	Whether an existing NCA enterprise data-modelling standard must be adopted.	Ch.2	SOURCE_DISCOVERY
6-A	Inventory of actual NCA/operator source systems, owners and provider channels.	Ch.6	SOURCE_DISCOVERY
7-A	Authoritative KPI set, formulas and units.	Ch.7	INSTITUTIONAL
12-A	Exact surcharge formula and charging basis; rate-change approval authority.	Ch.12	LEGAL
14-A	Exact lifecycle states counted toward SIM registration limits; field-level protection scope.	Ch.14	LEGAL

Decision ID	Issue	Affected Chapter	Category
15-A	Actual Anti-Fraud workflow, blocking interfaces and approval matrix at NCA.	Ch.15	SOURCE_DISCOVERY
16-A	NCA case/finding terminology and existing Regulatory Intelligence platform object model.	Ch.16	SOURCE_DISCOVERY
20-A	Retention period for ARGUS Interactions/Contributions; treatment of private vs shared interactions.	Ch.20	LEGAL
21-A	Room types, membership rules and default Participation Mode per Room type.	Ch.21	INSTITUTIONAL
26-A	Enterprise identity provider and existing NCA role model.	Ch.26	SOURCE_DISCOVERY
27-A	Audit retention period against applicable regulatory requirements.	Ch.27	LEGAL
31-A	Classification taxonomy and retention schedules under Ghanaian data-protection law.	Ch.31	LEGAL
32-A	Approved enterprise storage platforms and hosting constraints.	Ch.32	INSTITUTIONAL
36-A	Sovereignty/residency requirements and approved external/cloud processing, if any.	Ch.36	LEGAL
38-A	Acceptance authorities and required domain sign-offs for NCIE-003.	Ch.38	INSTITUTIONAL

42.4 Closure Process

A Decision ID closes only when its Required Evidence is actually supplied and recorded — an owner's verbal confirmation without recorded evidence does not close an item, consistent with this document's prohibition on converting SOURCE_DISCOVERY, INSTITUTIONAL or LEGAL items into invented facts. Closing a Decision ID updates its originating chapter's Human Review Focus entry (from open to resolved) in the next revision.

42.5 Prioritisation Before Field-Level Expansion

Decisions blocking Chapter 39's eventual field-level population — principally the SOURCE_DISCOVERY items in §42.3 — should be prioritised before further per-source dictionary expansion proceeds, since populating Source-System Field Mapping columns without them would require exactly the fabrication this document's instructions prohibit.

NCIE-002 Architecture Dependency & Traceability

This register consolidates items already individually traced to NCIE-002 within their originating chapters (see Ch.41 §41.2).

42.6 Acceptance Criteria

- Every INSTITUTIONAL, SOURCE_DISCOVERY and LEGAL item raised in Chapters 1-40 is represented in this register, directly or by cross-reference.
- No Decision ID is marked closed without a recorded Required Evidence entry.

NCIE-003 Acceptance Checklist & Handover to NCIE-004

Defines the formal gate for accepting NCIE-003 and handing its canonical data contracts to the next NCIE artifact.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

43.1 Purpose

Defines the formal gate for accepting NCIE-003 and handing its canonical data contracts to NCIE-004 (implementation-level data engineering) and downstream specifications.

43.2 Principal Entities

Table 60. Acceptance Checklist & Handover principal entities.

Entity	Attributes / Attribute Classes
Acceptance Checklist	item, status, evidence reference.
Open-Issue Status	reference to Ch.42's register, count by category, blocking vs non-blocking.
Traceability Completeness	confirmation that Ch.41 §41.2 covers every chapter.
Dictionary Completeness	confirmation that Ch.39's 15 mandated domains are represented and SOURCE DISCOVERY REQUIRED items are tracked, not fabricated.
ERD Consistency	confirmation that Ch.40 agrees with Ch.39 (§40.16, §40.18).
Approval Record	approving authority, date, scope of approval.
Baseline Version	the specific NCIE-003 version being accepted.
Handover Package	contents list, recipient (NCIE-004/Codex), exclusions (e.g. no fabricated source schemas).

43.3 Acceptance Checklist

- No material semantic conflict remains unresolved at baseline (verified against Ch.3 §3.7's and Ch.19/20/21's mandatory semantic separations).
- Data Dictionary (Ch.39), ERDs (Ch.40) and Traceability (Ch.41) agree with one another.
- Every INSTITUTIONAL/LEGAL/SOURCE_DISCOVERY item in Ch.42 has an assigned owner, even if not yet closed.
- No chapter fabricates a source-system schema, field name or enumeration claimed to be currently used by NCA (Ch.39 §39.2, throughout).
- Every chapter carries its classified Human Review Focus section (Ch.1-Ch.38).
- The surcharge rate and SIM registration limits are represented exclusively via Ch.24's governed Rule structures, with no literal constant elsewhere (verified against Ch.12 §12.4, Ch.14 §14.4).
- ARGUS-related entities preserve AI origin, human modification, and exclude private chain-of-thought (Ch.20 §20.2, verified against Ch.39 §39.12).

43.4 What NCIE-004 Consumes

NCIE-004 (the next artifact in the suite) receives: the full Logical Canonical Model (Ch.3-Ch.30), the Data Dictionary structure and worked entities (Ch.39), the logical ERDs (Ch.40), the NCIE-002 crosswalk (Ch.41), and the open decisions register (Ch.42) as its own starting backlog of discovery work. NCIE-004 does not receive, and must not fabricate in its own right, any physical schema this document deliberately deferred (Ch.32, Ch.40 §40.17).

43.5 Baseline-Lock and Change Control

Once accepted, the NCIE-003 baseline is locked under the same change-control process defined in Chapter 1 §1.4 — a canonical definition change after acceptance follows proposal, impact assessment (including against Ch.41's crosswalk), approval, version increment and a retained change record.

NCIE-002 Architecture Dependency & Traceability

This chapter is the formal closure of the authority chain stated in this edition's front matter §0.1: NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data semantics, ready for NCIE-004.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm NCIE-003 approvers and their scope of authority.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm precisely what NCIE-004 will consume from NCIE-003 versus re-derive independently.
- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the baseline-lock and change-control procedure to apply after acceptance.

43.6 Acceptance Criteria

- Every item in §43.3's checklist is demonstrably satisfied or explicitly logged as an open item in Ch.42 before Baseline Version status changes from IN DEVELOPMENT to Approved.
- The Handover Package is assembled with all §43.4 contents present and no excluded physical/fabricated content included.